

THE DISTRIBUTION OF THE LARGEST NEW EIGENVALUES OF RANDOM COVERS OF GRAPHS

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ABSTRACT. For a connected base graph B , the eigenvalues of a random degree k cover of B split into those of B and $(k-1)|V(B)|$ new eigenvalues, and the cover is *one-sided Ramanujan* if every new eigenvalue is at most the spectral radius $\rho(B)$ of the universal cover of B . Analyzing 1.4×10^8 samples of the largest new eigenvalue across ten families of random covers, we conjecture that its fluctuations follow the Tracy-Widom law of the symmetry class $\beta \in \{1, 2, 4\}$ selected by the Frobenius-Schur type of the associated irreducible representation of the covering group, centered at $\rho(B)$ and on a scale fixed by the square root edge of the spectral measure of the universal cover, so that the limiting probability that a random cover is Ramanujan is

$$\prod_{[\rho] \neq 1} F_{\beta(\rho)}(0),$$

a product over the non-trivial representations up to conjugation. With nothing fitted, every family matches the predicted scale to within 1.2% at its largest size, except for the slowest base, where it is within 4.5% and still converging. The leading finite size correction is a shift of order $1/n$ with a model dependent constant, which makes the Ramanujan probability converge only like $n^{-1/3}$ and accounts for the approximately 52% conjectured by Miller, Novikoff, and Sabelli for random regular graphs, where the limit $F_1(0) = 0.8319081 \dots$ has since been proven by Huang, McKenzie, and Yau. The far tails consist of localized eigenvalues of rare defect subgraphs, whose values and rates we compute and match to the data.

1. INTRODUCTION

A degree k cover (or k -lift) H of a connected base graph B is obtained by assigning to each edge of B a permutation in S_k , and the eigenvalues of H split into the *old* eigenvalues, which are those of B , and the remaining $(k-1)|V(B)|$ *new* eigenvalues. Random covers were introduced as a combinatorial construction of expanders by Amit and Linial [10, 11] (for expanders see the survey [57]), and the extremal behavior of their new eigenvalues is governed by the spectral radius $\rho(B)$ of the universal cover of B : Friedman [37] formulated the relative Alon conjecture, that the largest new eigenvalue of a random degree k cover of B is at most $\rho(B) + \epsilon$ asymptotically almost surely, successive bounds were given by Linial and Puder [71], by Lubetzky, Sudakov, and Vu [73], and by Puder [92], and the conjecture was proven for regular base graphs by Friedman and Kohler [42, 41] and in general by Bordenave [16] (see also [18, 19]). A sharp quantitative form was given recently by Chen, Garza-Vargas, Tropp, and van Handel [27, 28], and we refer to the surveys [103, 104, 17] for the strong convergence framework behind these results. With this in mind, we call a cover H of B *one-sided Ramanujan* if every new eigenvalue is at most $\rho(B)$. When B is d -regular the universal cover is the d -regular

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tree and $\rho(B) = 2\sqrt{d-1}$, the threshold of the classical Ramanujan property, and no infinite family does better, by the Alon-Boppana bound (see Nilli [87, 88] and Friedman [36]). When B is the bouquet of $d/2$ loops on a single vertex, a random degree k cover of B is precisely the permutation model of a random d -regular graph on k vertices, with new eigenvalues its non-trivial eigenvalues.

Covers also carry the constructive side of the Ramanujan story. Lubotzky, Phillips, and Sarnak [74] and independently Margulis [79] built infinite Ramanujan families when $d-1$ is prime, and whether non-bipartite Ramanujan graphs exist for every degree remains open, though explicit near Ramanujan graphs of every degree and size are known [4]. Bilu and Linial [15] conjectured that every d -regular graph has a 2-cover whose new eigenvalues all lie in $[-2\sqrt{d-1}, 2\sqrt{d-1}]$, which for Ramanujan base graphs would settle it. A counterexample among general cubic graphs has recently been announced by Xu [106], while the case of Ramanujan base graphs, which would suffice, remains open. Marcus, Spielman, and Srivastava [78] proved the one-sided statement for bipartite graphs, and Hall, Puder, and Sawin [49] proved that every connected loopless graph has a one-sided Ramanujan cover of every degree. For the group constrained covers studied below, Chandrasekaran and Velingker [26] constructed shift lifts of degrees three and four preserving the Ramanujan property, and Agarwal, Chandrasekaran, Kolla, and Madan [2] bounded the new eigenvalues of random group based lifts, showing in particular that abelian lifts of exponentially large degree are never close to Ramanujan.

The relative Alon theorem localizes the largest new eigenvalue at the threshold but says nothing about which side of it the eigenvalue falls on, and the question of this paper is the finer one: how does the largest new eigenvalue of a random cover fluctuate around $\rho(B)$, and what proportion of random covers is Ramanujan? At the bouquet this is a question with history. Alon [3] conjectured that random d -regular graphs are nearly Ramanujan, Friedman, Kahn, and Szemerédi [40, 35] proved the first $O(\sqrt{d})$ bounds, Friedman [38] proved the conjecture, and Sarnak [96] asked whether a positive proportion is Ramanujan outright, reporting experiments of Novikoff [89] already pointing to the Tracy-Widom law. The first systematic numerical attack was made in 2008 by Miller, Novikoff, and Sabelli [83], hereafter MNS. Writing $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_V$ for the adjacency eigenvalues of a d -regular graph on V vertices, they sampled 1000 graphs per size from bipartite and perfect matching models with $d \in \{3, 4, 7, 10\}$ and $V \leq 20,000$, with 100,000 graphs at three selected sizes, and, motivated by the numerical evidence of Gaussian Orthogonal Ensemble spacing statistics in the bulk found by Jakobson, Miller, Rivin, and Rudnick [63], gave evidence that the fluctuations of λ_2 follow the $\beta = 1$ Tracy-Widom law of random matrix theory [102]. Writing μ_V and σ_V for the mean and standard deviation of λ_2 over graphs of size V , they fitted power laws

$$2\sqrt{d-1} - \mu_V \sim c_\mu V^m, \quad \sigma_V \sim c_\sigma V^s,$$

found best fit exponents m between -0.72 and -0.83 against s between -0.71 and -0.73 , and were led to conjecture that the limiting probability of being one-sided Ramanujan is the mass of the Tracy-Widom law to the left of its mean,

$$(1.1) \quad F_1(\mu_{TW_1}) = 0.519652\dots,$$

approximately 52%. From a formal expansion of the graph zeta function, Friedman [39] later suggested that the limiting probability exceeds one half, for random regular graphs and for random covers of a regular base alike, without identifying the constant.

For degrees growing with the number of vertices, Bauerschmidt, Huang, Knowles, and Yau [14] proved Tracy-Widom fluctuations for $V^{2/9+\epsilon} \leq d \leq V^{1/3-\epsilon}$, He [50] for

$V^{2/3+\epsilon} \leq d \leq V/2$, and Huang and Yau [61] for the full sparse regime $V^\epsilon \leq d \leq V^{1/3-\epsilon}$, concluding that about 69% of such graphs are Ramanujan and correcting the prediction (1.1): the correct one-sided constant is 83%, not 52%. For fixed degree, where Huang and Yau [60] had located λ_2 within $V^{-\epsilon}$ of the threshold and optimal eigenvalue rigidity was then established in [58], Huang, McKenzie, and Yau [59] proved the definitive result (see also the lecture notes [62]).

Theorem 1. *Let $d \geq 3$ be fixed, let G be a uniformly random simple d -regular graph on V vertices, and let $A = d(d-1)/(d-2)^2$. Then*

$$(AV)^{2/3} \left(\frac{\lambda_2(G)}{\sqrt{d-1}} - 2 \right) \rightarrow TW_1$$

in distribution as $V \rightarrow \infty$, where TW_1 is the $\beta = 1$ Tracy-Widom distribution in its original normalization. In particular, since the centering is exactly $2\sqrt{d-1}$,

$$\lim_{V \rightarrow \infty} P \left(\lambda_2(G) \leq 2\sqrt{d-1} \right) = F_1(0) = 0.8319081 \dots$$

Unfolding the constants, Theorem 1 states that

$$(1.2) \quad \lambda_2(G) = 2\sqrt{d-1} + c(d)V^{-2/3}X_V, \quad c(d) = \left(\frac{(d-2)^2}{d(d-1)} \right)^{2/3} \sqrt{d-1},$$

where X_V converges in law to TW_1 , and so the mean and standard deviation of λ_2 are asymptotic to $|\mu_{TW_1}| c(d)V^{-2/3}$ below the threshold and $\sigma_{TW_1} c(d)V^{-2/3}$, where $\mu_{TW_1} = -1.2065335746 \dots$ and $\sigma_{TW_1} = 1.2679830577 \dots$ are the mean and standard deviation of the un-recentered Tracy-Widom law. Both of MNS's exponents equal $-2/3$, and the threshold sits $|\mu_{TW_1}|/\sigma_{TW_1} = 0.951537 \dots$ standard deviations to the right of the mean. The constant $c(d)$ has a transparent origin, which is the key to the constants of the conjectures below. The Kesten-McKay law, the limiting spectral measure of random d -regular graphs, has density

$$(1.3) \quad \frac{d\sqrt{4(d-1)-x^2}}{2\pi(d^2-x^2)} = \left(\frac{C(d)}{\pi} + o(1) \right) \sqrt{2\sqrt{d-1}-x}, \quad C(d) = \frac{d(d-1)^{1/4}}{(d-2)^2},$$

near its right edge, and $c(d) = C(d)^{-2/3}$. In other words, $(C(d)V)^{-2/3}$ is the distance from the edge within which a square root edge of density $(C(d)/\pi)\sqrt{2\sqrt{d-1}-x}$ holds a bounded number of V eigenvalues, and at this scale λ_2 fluctuates exactly as the largest eigenvalue of the Gaussian Orthogonal Ensemble does at its own edge, where the semicircle law on $[-2, 2]$ has $C = 1$.

For covers of a base graph with more than one vertex, the fluctuation theory is open, outside one result on iterated 2-covers of dense random regular graphs [72], and in this paper we report on a numerical study of it roughly a thousand times larger than the experiment of MNS. Our data supports a precise answer: the limiting fluctuations follow the Tracy-Widom law of the random matrix symmetry class dictated by the representation theory of the cover, on a scale predicted, with no free parameter, by the square root edge of the limiting spectral measure. When the covering permutations are constrained to a group Γ , the new spectrum decomposes into isotypic blocks indexed by the non-trivial irreducible representations of Γ (see section 2), and the Frobenius-Schur indicator of each representation sorts its block into one of the three classical ensembles: real representations give real symmetric blocks (GOE, $\beta = 1$), complex representations give complex Hermitian blocks (GUE, $\beta = 2$), and quaternionic representations give

quaternion self-dual blocks (GSE, $\beta = 4$), the threefold way of Dyson [34]. For bulk statistics this selection has a long history. Gamburd, Jakobson, and Sarnak [44] observed GOE and GSE level spacings in the real and quaternionic irreducible representations of random elements of the group ring of $SU(2)$, and proved the corresponding Gaussian limit as the number of generators grows, Gamburd, Lafferty, and Rockmore [45] observed GOE and GSE spacings in the irreducible components of quantized cat maps, and Rivin and Sardari [94] split the spectra of random Cayley graphs of $SL_2(\mathbb{Z}/p\mathbb{Z})$ by irreducible representation and conjectured that different representations behave independently. In quantum chaos, a time reversal invariant billiard with only a threefold symmetry shows GUE statistics [70, 68], the indicator governs the spacing statistics of desymmetrized spectra [66], and the quaternionic block of a quantum graph with the symmetry of the quaternion group shows GSE statistics without spin [67], which was then realized in a microwave experiment [93]. We find that the same selection governs the spectral edge, including the scale.

Conjecture 2. *Fix $d \geq 3$ and a finite group Γ , let ρ be a non-trivial irreducible representation of Γ of dimension d_ρ with Frobenius-Schur indicator $\varepsilon(\rho) \in \{1, 0, -1\}$, and let A_ρ denote the ρ -isotypic block of a uniformly random Γ -cover of a random simple d -regular base graph on n vertices, a Hermitian matrix of size $N_\rho \times N_\rho$ over $\mathbb{R}$, $\mathbb{C}$, or the quaternions $\mathbb{H}$ according as $\varepsilon(\rho) = 1, 0, -1$, where $N_\rho = nd_\rho$ for real and complex ρ and $N_\rho = nd_\rho/2$ for quaternionic ρ . Then*

$$\frac{\lambda_{\max}(A_\rho) - 2\sqrt{d-1}}{c(d) N_\rho^{-2/3}} \rightarrow \chi_\beta$$

in distribution as $n \rightarrow \infty$, where $\beta = 1, 2, 4$ according as $\varepsilon(\rho) = 1, 0, -1$, $c(d)$ is the constant of (1.2), and χ_β is the limit law of the rescaled largest eigenvalue of the Gaussian ensemble of class β with its semicircle law on $[-2, 2]$, namely

$$\chi_1 = TW_1, \quad \chi_2 = TW_2, \quad \chi_4 = 2^{-1/6}TW_4,$$

in the original normalization of Tracy and Widom. In particular the centering is exactly $2\sqrt{d-1}$, with no shift.

The factor $2^{-1/6}$ is not a new constant: it records the normalization Tracy and Widom chose for the GSE (see section 2.2). The largest new eigenvalue of the cover is the maximum of the block maxima, and blocks of representations that are not complex conjugates of one another behave independently in our data, as Rivin and Sardari conjectured for the bulk [94]. This yields explicit limiting Ramanujan probabilities.

Conjecture 3. *With the notation of Conjecture 2, the block maxima $\lambda_{\max}(A_\rho)$ are asymptotically independent across non-conjugate non-trivial representations, and the probability that a random Γ -cover is one-sided Ramanujan satisfies*

$$\lim_{n \rightarrow \infty} P\left(\lambda_{\max}(A_\rho) \leq 2\sqrt{d-1} \text{ for all } \rho\right) = \prod_{[\rho] \neq 1} F_{\beta(\rho)}(0),$$

where the product runs over the non-trivial irreducible representations of Γ up to complex conjugation, and F_β is the Tracy-Widom distribution function. Here $F_1(0) = 0.8319081\dots$, $F_2(0) = 0.96937\dots$, and $F_4(0) = 0.99857\dots$.

When all the non-trivial blocks share one symmetry class the largest new eigenvalue itself is Tracy-Widom, and this is what the data shows: $\mathbb{Z}_3$ covers follow χ_2 , quaternion covers follow χ_4 , and permutation covers, whose new spectrum is the single block of

the real standard representation of S_k , follow χ_1 . For every family over a regular base the scale is that of Conjecture 2 to within one percent at the largest sizes, with nothing fitted (Figure 3.7). When the blocks mix classes no single Tracy-Widom law can appear, and the data confirms this too: the largest new eigenvalue of a $\mathbb{Z}_4$ cover, a maximum of one GOE block and one GUE block, and of a $\mathbb{Z}_5$ cover, a maximum of two independent GUE blocks, each select the predicted maximum law over every Tracy-Widom law (Figure 4.1), and the $\mathbb{Z}_5$ family matches its Ramanujan probability $F_2(0)^2$ to three decimal places.

The conjectures have a hyperbolic counterpart. Random degree k covers of a fixed hyperbolic surface are known to have relative spectral gap $\frac{3}{16} - \epsilon$ [75] and in fact $\frac{1}{4} - \epsilon$ [54, 76], with a polynomial rate [52], the analog of the relative Alon theorem. For smooth linear spectral statistics of the Laplacian twisted by a character of the fundamental group, Naud [86] proved that the variance is that of the GOE for a real character and of the GUE otherwise, with extensions to higher dimensional twists [86, 85, 77], the GSE appearing for $SU(2)$ twists [85], while for Weil-Petersson random surfaces the GOE, GUE, and GSE statistics appear for the Laplacian, for twisted Laplacians, and for Dirac operators [95, 80]. At the bottom of the spectrum, the $\beta = 1$ Tracy-Widom law has been conjectured for the first new eigenvalue of random covers and of Weil-Petersson surfaces [52, 53], and the selection of β by the type of the twist that Conjecture 2 asserts for graphs has, to our knowledge, no analog there yet.

Our final conjecture extends the picture to covers of a fixed irregular base graph, where no theorem and, to our knowledge, no prior experiment exists. For regular bases, Lubetzky, Sudakov, and Vu [73] simulated random covers of three cubic Ramanujan graphs and conjectured that the probability that a random cover is Ramanujan converges to a constant in $(0, 1)$, and the following conjecture identifies their constant, universally in the base graph. For a finite graph B on n vertices, let μ_B be the average over the vertices v of B of the spectral measure of the adjacency operator of the universal cover at a lift of v . This is the limiting spectral measure of the new eigenvalues of random covers of B , it is the Kesten-McKay law when B is regular, and in each of our examples its density vanishes like a square root at $\rho(B)$.

Conjecture 4. *Let B be a fixed connected base graph on n vertices of cycle rank $e(B) - v(B) + 1$ at least two, let $\rho(B)$ denote the spectral radius of its universal cover, and suppose that the density of μ_B has a square root edge at $\rho(B)$, with the constant $C_B > 0$ defined by*

$$\frac{d\mu_B}{dx} = \left(\frac{C_B}{\pi} + o(1) \right) \sqrt{\rho(B) - x} \quad \text{as } x \rightarrow \rho(B)^-.$$

Let H_k be a uniformly random degree k cover of B , and let $\lambda_{\text{new}}(H_k)$ be its largest new eigenvalue. Then

$$(C_B nk)^{2/3} (\lambda_{\text{new}}(H_k) - \rho(B)) \rightarrow TW_1$$

in distribution as $k \rightarrow \infty$. In particular

$$\lim_{k \rightarrow \infty} P(H_k \text{ is one-sided Ramanujan}) = F_1(0) = 0.8319081 \dots$$

for every such base graph B .

The $\beta = 1$ law in Conjecture 4 is forced by the picture above, since the standard representation of S_k is real for every k , and the scale is the square root edge scale of μ_B with nk eigenvalues, which for the bouquet of $d/2$ loops, where $n = 1$ and $C_B = C(d)$, is the scale of Theorem 1. The hypothesis excludes only the degenerate bases: covers of a

tree are disjoint copies of it and covers of a cycle are unions of longer cycles, so neither has a fluctuation theory, while vertices of degree two are harmless, since covers of a subdivision are subdivisions of covers. Conjecture 4 is thus the common generalization of the proven regular case to arbitrary base graphs. We test it in section 5 at five million covers per size of the fixed bases K_5 and $K_5 - e$: each family selects the TW_1 law, the empirical Ramanujan probabilities at $V = 10,000$ are 0.83281 and 0.82916 against the conjectured $F_1(0) = 0.8319081\dots$, and the fitted amplitudes extrapolate to within 0.2% of the predicted $C_B^{-2/3}$, which for $K_5 - e$ is the new constant 0.695228... computed from the universal cover. A third fixed base, $K_4 - e$, tests the conjecture at its boundary, cycle rank exactly two, where the square root edge of μ_B is narrow, the defect structure of the far tail changes, and the convergence is slowest. The far tails themselves are not noise: they are localized eigenvalues of polynomially rare defect subgraphs, computable exactly from the universal cover, and in section 5 we prove that every such defect occurs, and produces its eigenvalue, at the polynomial rate governed by its cycle rank (Propositions 5 and 6), conjecture that the defects account for the entire tail, and verify the predicted values and rates in the data.

The data also shows how the limits are approached, which no theorem addresses. In every family the finite size law is a shift of the whole law by a/n , where n is the size and a is a model dependent constant, together with a scale which converges to its predicted value. For the random 4-regular graphs conditioned to be simple the shift dominates, with $a = 4.465 \pm 0.008$, and so the Ramanujan probability at size V is $F_1(a/(c(4)V^{1/3}))$ to within 3×10^{-4} for every V from 1000 to 20,000, a function which approaches $F_1(0)$ from above only like $V^{-1/3}$, and is still 0.01 above it at $V = 8.6 \times 10^5$. This one constant also reproduces the drift of the exponents that MNS fitted (section 3.3), and so it is the origin of their 52%. For the permutation model without the conditioning, the multigraph, $a = -1.44 \pm 0.10$: the law of its bulk is the same, and its apparent departure from Tracy-Widom, visible in every statistic built from the sample moments, comes from a thin tail of defect eigenvalues of cycle rank two, at $7/2$ and at the real root of $\lambda^3 = 8\lambda + 16$, which occur at rate $1/V$ and dominate the moments (section 3.4).

Conjectures 2, 3, and 4, without the constants $c(d)N_\rho^{-2/3}$ and $(C_{Bnk})^{-2/3}$, date to 2018 and were formulated jointly with Lior Silberman, following a suggestion of Peter Sarnak. All of the data reported here was generated by 2023, before the appearance of [59], with every family but the fixed base ones predating [61] as well, and so the pipeline was in no way tuned to the theorems it now confirms. The regular graph data of section 3 thus serves as a control: the pipeline that recovers the proven law and its constants, with nothing fitted, is the pipeline that carries the cover results. It also measures what the theorems do not address, namely the finite size law, the diagnosis of MNS's 52%, and the multigraph model, while the cover families of sections 4 and 5 are the substance of the paper.

We stress three differences between our experiment and that of MNS. First, scale: we use 5×10^6 samples per size against their 1000, sizes up to $V = 500,000$ against their 20,000, and 1.4×10^8 samples in total, so that the four candidate laws, whose distribution functions differ only in the third decimal, are separated by dozens of binomial standard errors rather than fractions of one (Figure 2.4). Second, methodology: instead of fitting power laws to the moments we compare each sample with its conjectured law directly, with the centering and scale fixed in advance, and separate the finite size corrections into a location, a scale, and a shape, which reveals both the $1/n$ shift and why a global power law fit over $V \leq 20,000$ overestimates the mean exponent. Third, and principally, scope:

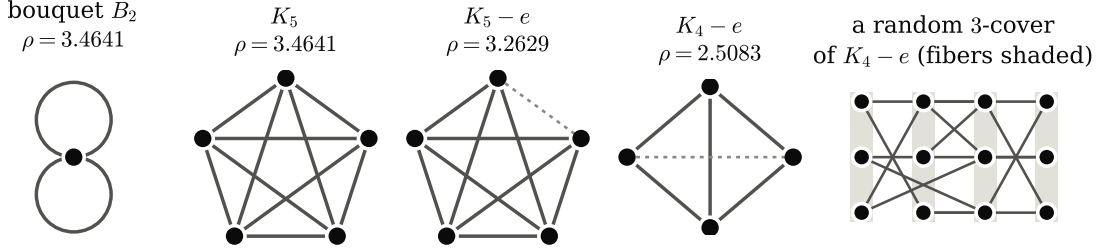


FIGURE 2.1. The base graphs of the dataset, with the spectral radius ρ of each universal cover: the bouquet of two loops, whose random covers are the permutation model of random 4-regular graphs, the complete graph K_5 , and the irregular graphs $K_5 - e$ and $K_4 - e$ (the removed edge dotted). Right: a random 3-cover of $K_4 - e$, drawn with the fiber over each base vertex as a column.

the nearest precursors to our cover experiments are the simulation of [73], one thousand covers per size of three cubic Ramanujan bases, which observed a stable Ramanujan proportion without identifying a law, and the per representation experiments of [94] on Cayley graphs. The Tracy-Widom identification at the edge, its selection by the representation type, the scale constants, and the Ramanujan probabilities have, to our knowledge, no analog in any published experiment. Throughout, we study the one-sided question only, and make no claims about the joint behavior of the smallest new eigenvalue.

1.1. Outline. In section 2 we describe the random cover model, the isotypic decomposition, the limit laws and their scale constants, the statistics we compute, and the dataset. In section 3 we present the regular graph control experiment: the parameter-free convergence to Theorem 1, the finite size law and its $1/V$ shift, the diagnosis of MNS's conjecture, and the multigraph model. Section 4 presents the evidence for Conjectures 2 and 3 across abelian and quaternion covers, including the scale constants and the mixed class families. Section 5 presents the data for Conjecture 4 on the three fixed base families, the universal cover prediction of their amplitudes, and the defect analysis of the far tails, including the two propositions and the completeness conjecture that organize them. Section 6 collects open problems.

2. THE NUMERICAL MODEL AND THE DATASET

2.1. Random Covers and the Isotypic Decomposition. Let B be a connected base graph on n vertices, possibly with loops and multiple edges. A degree k cover of B is determined by a voltage assignment σ giving each oriented edge $e = (u, v)$ of B a permutation $\sigma_e \in S_k$, with $\sigma_{\bar{e}} = \sigma_e^{-1}$ on the reversed edge: the cover H has vertex set $V(B) \times \{1, \dots, k\}$, with (u, i) adjacent to $(v, \sigma_e(i))$ for each edge $e = (u, v)$. A *random cover* of degree k assigns independent uniform permutations to the edges of B . More generally, given a finite group Γ of order k , a random Γ -cover assigns independent uniform elements of Γ to the edges, acting on the fiber Γ by right multiplication. The eigenvalues of B lift to H , and are called the old eigenvalues, and the remaining $n(k-1)$ eigenvalues are called *new*, and we write $\lambda_{\text{new}}(H)$ for the largest new eigenvalue. Figure 2.1 shows the four base graphs of our dataset and a random cover.

The spectrum of a Γ -cover organizes along the irreducible representations of Γ . Decomposing the permutation action of Γ on its group algebra into irreducibles, the adjacency matrix of H is conjugate to a direct sum

$$(2.1) \quad A_H \cong \bigoplus_{\rho \in \text{Irr}(\Gamma)} A_\rho^{\oplus d_\rho}, \quad A_\rho = \sum_{e=(u,v)} E_{uv} \otimes \rho(\sigma_e),$$

where the sum runs over the oriented edges of B , E_{uv} is the matrix unit, and d_ρ is the dimension of ρ , so that A_ρ is an nd_ρ dimensional Hermitian matrix (see [49, Section 2] for this decomposition, [2] for its use on group based lifts, and [13, 99] for its counterparts for quantum graphs and graph zeta functions). The trivial representation recovers the base, $A_1 = A_B$, and so the new spectrum of H is the union of the spectra of the non-trivial blocks A_ρ , each with multiplicity d_ρ . Note that the blocks of complex conjugate representations satisfy $A_{\bar{\rho}} = \overline{A_\rho}$ and have identical spectra, and so the distinct sources of randomness are indexed by the non-trivial representations up to conjugation. For a full symmetric cover the same decomposition applies with $\Gamma = S_k$ acting on $\{1, \dots, k\}$: the permutation representation splits as the trivial plus the standard $(k-1)$ dimensional representation, and the new spectrum is the single block of the standard representation.

The Frobenius-Schur indicator

$$\varepsilon(\rho) = \frac{1}{|\Gamma|} \sum_{g \in \Gamma} \chi_\rho(g^2) \in \{1, 0, -1\}$$

sorts the irreducible representations into real ($\varepsilon = 1$), complex ($\varepsilon = 0$), and quaternionic ($\varepsilon = -1$) types, and correspondingly sorts the blocks A_ρ into the three classical matrix symmetry classes: a real representation may be realized over $\mathbb{R}$, making A_ρ conjugate to a real symmetric matrix, a complex representation gives a complex Hermitian block with no further symmetry, and a quaternionic representation gives a self-dual block, an $nd_\rho/2$ dimensional Hermitian matrix over $\mathbb{H}$, whose eigenvalues carry an even multiplicity. In random matrix theory these are the symmetry classes of the GOE, GUE, and GSE, whose largest eigenvalues fluctuate according to the Tracy-Widom laws [101, 102]. Our families realize all three types: for $\Gamma = \mathbb{Z}_k$ the non-trivial characters $\chi_j(x) = e^{2\pi i j x/k}$ give n dimensional blocks with entries $\chi_j(\sigma_e)$ on the edges of the base, complex Hermitian in general and real symmetric exactly when $\chi_j = \pm 1$ is real valued, while the quaternion group Q_8 has a unique non-trivial higher dimensional representation, the two dimensional representation of quaternionic type, whose real form is four dimensional. As k grows, the $\mathbb{Z}_k$ blocks approach the regular graphs with independent uniform phases on the edges studied by Oren, Godel, and Smilansky [90, 91], who compared their bulk statistics with the GUE.

Concretely, $\mathbb{Z}_3$ has one conjugate pair of complex characters, and so the largest new eigenvalue of a random $\mathbb{Z}_3$ cover is the largest eigenvalue of a single GUE class block. $\mathbb{Z}_4$ has the real character $\chi_2 = (-1)^x$ together with one conjugate pair, and so its largest new eigenvalue is the maximum of a GOE class and a GUE class maximum. $\mathbb{Z}_5$ has two conjugate pairs, giving the maximum of two GUE class maxima with identical marginal distributions. In general, the non-trivial characters of $\mathbb{Z}_k$ form $\lfloor (k-1)/2 \rfloor$ conjugate pairs, together with the real character $\chi_{k/2}(x) = (-1)^x$ when k is even, and so the largest new eigenvalue of a random cyclic cover is the maximum of $\lfloor (k-1)/2 \rfloor$ independent GUE class maxima and, for even k , one GOE class maximum, with limiting Ramanujan probability

$$F_2(0)^{\frac{k-1}{2}} \quad \text{for odd } k, \quad F_1(0) F_2(0)^{\frac{k-2}{2}} \quad \text{for even } k$$

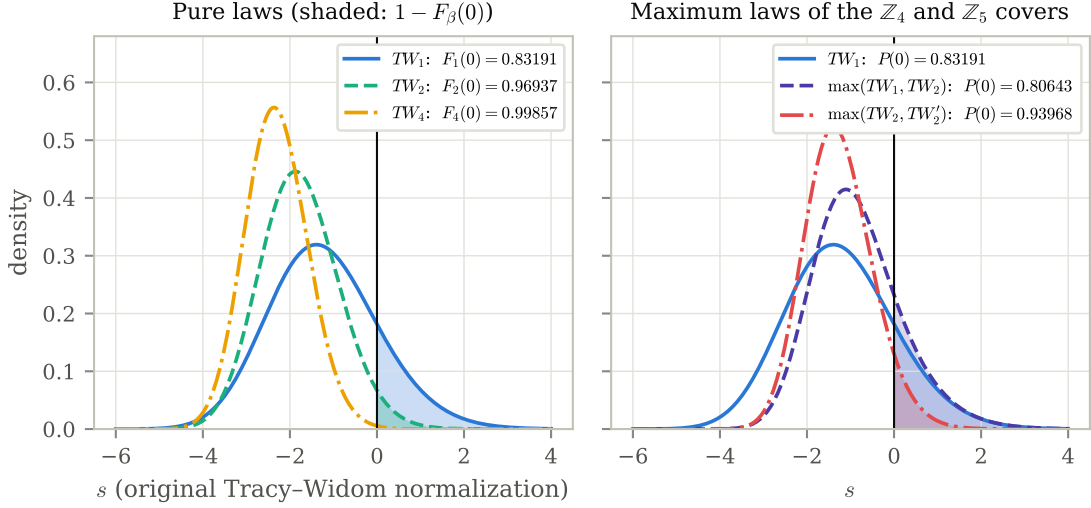


FIGURE 2.2. Left: the densities of TW_1 , TW_2 , and TW_4 in the original normalization, with the mass $1 - F_\beta(0)$ to the right of the origin shaded, the limiting probability that a block of class β is not Ramanujan. Right: the densities of the maximum laws of the $\mathbb{Z}_4$ and $\mathbb{Z}_5$ covers, against TW_1 .

under Conjecture 3. The quaternion family samples the four dimensional real form of the quaternionic block of a random Q_8 -cover directly, a $4n$ dimensional matrix all of whose eigenvalues are new, with each distinct eigenvalue repeated by the quaternionic structure, and so $N_\rho = n$. These five families, together with the permutation covers of the bouquet (random 4-regular graphs), of K_5 , of K_5 minus an edge, and of K_4 minus an edge, are the subject of sections 3 through 5.

2.2. The Limit Laws and the Scale Constants. By TW_β we mean the Tracy-Widom law in the original normalization of Tracy and Widom [101, 102], with distribution function F_β , mean μ_{TW_β} , and standard deviation σ_{TW_β} , and by $\widetilde{TW}_\beta$ the standardization to mean 0 and variance 1. The constants we use are

	β	μ_{TW_β}	σ_{TW_β}	$ \mu_{TW_\beta} /\sigma_{TW_\beta}$	$F_\beta(\mu_{TW_\beta})$
(2.2)	1	-1.2065335746	1.2679830577	0.9515376	0.519652
	2	-1.7710868074	0.9017731382	1.9640048	0.515016
	4	-2.3068848932	0.7195302084	3.2060987	0.511069

following [21, 20], and the values $F_1(0) = 0.8319081\dots$, $F_2(0) = 0.96937\dots$, $F_4(0) = 0.99857\dots$ of the distribution functions at the origin. We evaluate F_β as Fredholm determinants by Gauss-Legendre quadrature, following Bornemann [21], which is accurate to near machine precision and so usable deep into both tails, and which reproduces every constant in (2.2) to the digits shown. Figure 2.2 shows the three densities, and the maximum laws of section 4.3.

The scales in Conjecture 2 refer to the Gaussian ensembles normalized alike. Let $N^{2/3}(\lambda_{\max} - 2) \rightarrow \chi_\beta$ for the Gaussian ensemble of class β and dimension N over $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$, normalized so that its semicircle law is supported on $[-2, 2]$. For $\beta = 1, 2$ this is the definition of TW_β , while for $\beta = 4$ Tracy and Widom normalized the GSE differently [102, Section VII]. In the notation of Bornemann's review, the soft edge limit of the GSE of complex dimension $2N$ is taken at $\sqrt{4N}$ on the scale $2^{-1/2}(2N)^{-1/6}$ [20, (2.4)], and $F_4(s)$ is its distribution function evaluated at $\sqrt{2}s$ [20, (2.10)]. Rescaling

the edge $\sqrt{4N}$ to 2 gives

$$\chi_4 = 2^{-2/3} \sqrt{2} TW_4 = 2^{-1/6} TW_4.$$

A direct simulation of 2×10^4 samples of this ensemble with $N = 200$ gives a standard deviation of 0.6417 for $N^{2/3}(\lambda_{\max} - 2)$, against $2^{-1/6} \sigma_{TW_4} = 0.6410$, while the same simulation of the GOE and GUE gives 1.2620 and 0.9054, against $\sigma_{TW_1} = 1.2680$ and $\sigma_{TW_2} = 0.9018$.

The scale constants themselves come from the square root edge of the limiting spectral measure, as in (1.3). For a base graph B , the measure μ_B of Conjecture 4 is computable from the branch values $y_{u \rightarrow v}(z)$, the solution with positive imaginary part for $\text{Im } z > 0$ of

$$(2.3) \quad y_{u \rightarrow v}(z) = \left(z - \sum_{w \sim v, w \neq u} y_{v \rightarrow w}(z) \right)^{-1}$$

over the directed edges of B , since the Green's function of the universal cover at a lift of v is $G_v(z) = (z - \sum_{w \sim v} y_{v \rightarrow w}(z))^{-1}$ and $d\mu_B/dx = -\frac{1}{\pi n} \sum_v \text{Im } G_v(x + i0)$. Solving (2.3) by Newton's method, continued from the upper half plane to within 10^{-12} of the real axis, and extrapolating $\pi(d\mu_B/dx)/\sqrt{\rho(B) - x}$ to the edge (Figure 2.3), we obtain

$$(2.4) \quad \begin{array}{cccc} B & \rho(B) & C_B & C_B^{-2/3} \\ K_5 & 2\sqrt{3} = 3.4641016 & 3^{1/4} = 1.3160740 & 3^{-1/6} = 0.8326832 \\ K_5 - e & 3.2628765 & 1.7250777 & 0.6952285 \\ K_4 - e & \sqrt{1 + 2\sqrt{7}} = 2.5082868 & 10.2071395 & 0.2125188 \end{array}$$

where the K_5 row recovers $C(4)$ and $c(4)$ of (1.2) and (1.3). By Conjecture 4, the largest new eigenvalue of a random cover of B on $V = nk$ vertices is $\rho(B) + C_B^{-2/3} V^{-2/3} X$ with X asymptotically TW_1 . Note that the square root law for $K_4 - e$ sets in only within about 10^{-3} of its edge, against 10^{-2} for $K_5 - e$, which will matter in section 5.3.

2.3. The Statistics. For each family and size we record the sample mean μ_V , standard deviation σ_V , skewness, and excess kurtosis of the largest new eigenvalue λ , the Kolmogorov-Smirnov distance between the standardized sample and each of $\widetilde{TW}_1, \widetilde{TW}_2, \widetilde{TW}_4$ and the standard normal, the fraction of samples below the Ramanujan threshold, and the fraction below the sample mean. The last statistic is MNS's discriminator: the standardized Tracy-Widom laws are nearly indistinguishable by their overall shape, however their masses to the left of the mean, listed in (2.2), differ in the third decimal, and with 10^6 to 5×10^6 samples the binomial standard error is between 2×10^{-4} and 5×10^{-4} , separating the four candidates by tens of standard errors. At MNS's 1000 samples per size the same separation is a third of a standard error (Figure 2.4).

The relevance of the ratio $g_V = (\rho - \mu_V)/\sigma_V$ of the mean gap to the standard deviation, with ρ the family's threshold, is the following elementary observation of MNS. For a d -regular family, the probability of being one-sided Ramanujan is

$$P\left(\lambda \leq 2\sqrt{d-1}\right) = P\left(\frac{\lambda - \mu_V}{\sigma_V} \leq g_V\right),$$

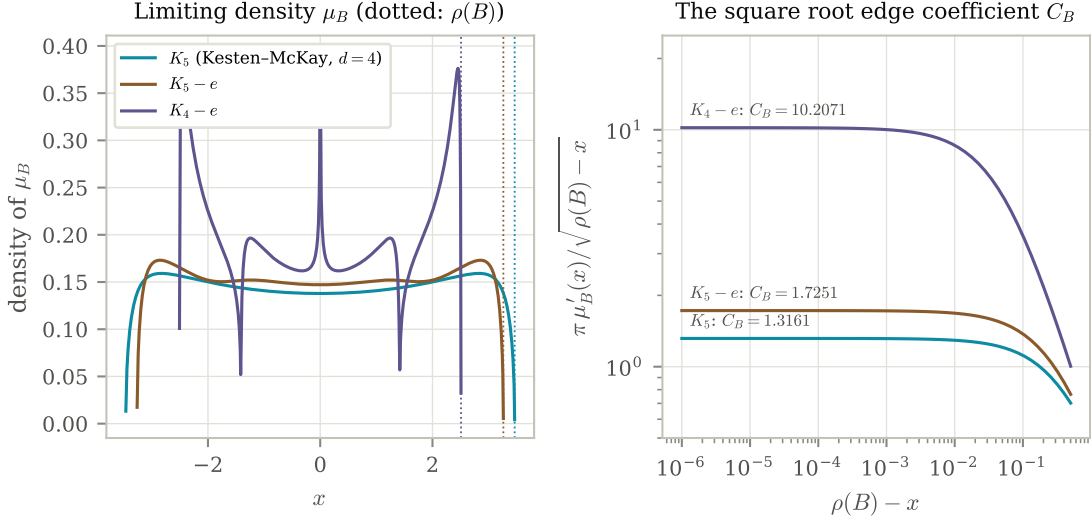


FIGURE 2.3. Left: the limiting densities μ_B of the new eigenvalues of random covers of K_5 (the Kesten-McKay law with $d = 4$), $K_5 - e$, and $K_4 - e$, computed from the branch recursion (2.3), with the edges $\rho(B)$ dotted. Right: $\pi \mu'_B(x) / \sqrt{\rho(B) - x}$ against the distance to the edge, which converges to the constant C_B of (2.4). The square root regime of $K_4 - e$ is an order of magnitude narrower than those of K_5 and $K_5 - e$.

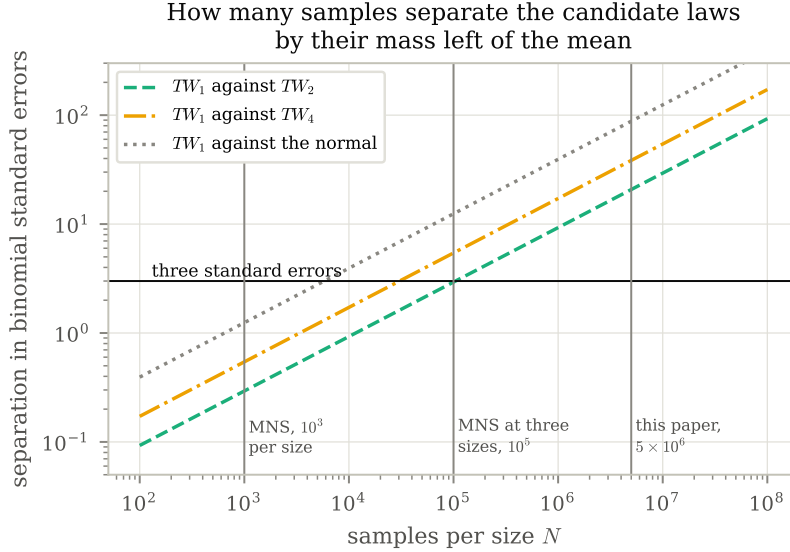


FIGURE 2.4. The separation, in binomial standard errors, of the mass of TW_1 to the left of its mean from those of TW_2 , TW_4 , and the normal law, against the number N of samples per size. With $N = 1000$, as in most of MNS's sizes, TW_1 and TW_2 are separated by 0.3 standard errors, and with our 5×10^6 by 21.

and so if the standardized law converges to $\widetilde{TW}_1$, then

$$(2.5) \quad \lim_{V \rightarrow \infty} P\left(\lambda \leq 2\sqrt{d-1}\right) = \begin{cases} F_1(\mu_{TW_1}) = 0.519652\dots & \text{if } g_V \rightarrow 0 \\ \widetilde{F}_1(r) & \text{if } g_V \rightarrow r \in (0, \infty) \\ 1 & \text{if } g_V \rightarrow \infty \end{cases}.$$

MNS's power law fits pointed to the first case. The Tracy-Widom scaling (1.2), with no shift in the centering, forces the middle case with $r = |\mu_{TW_1}|/\sigma_{TW_1}$, where $\bar{F}_1(r) = F_1(0)$, and this is what our data shows and what Theorem 1 now proves.

Standardizing by the sample mean and standard deviation discards the two numbers the conjectures predict, and so we also compare each sample with its conjectured law directly. For a family whose conjectured law is $\lambda \approx \rho + cN^{-2/3}\chi$, where N is the block dimension N_ρ of Conjecture 2 ($N = V$ for the regular and fixed base families, and $N = n$ for the character and quaternion blocks over a base on n vertices), we form the *parameter-free* variable

$$s = \frac{(\lambda - \rho)N^{2/3}}{c},$$

with $c = c(4)$ for every 4-regular family and $c = C_B^{-2/3}$ for the fixed bases, and nothing fitted. To separate the finite size corrections, we regress the empirical quantiles s_p on the quantiles χ_p of the conjectured law over $0.05 \leq p \leq 0.95$,

$$s_p \approx \alpha_N + \gamma_N \chi_p,$$

and report the *scale ratio* γ_N , which should tend to 1, and the *location coefficient* $a_N = -\alpha_N c N^{1/3}$, defined so that the fitted law of λ is

$$(2.6) \quad \lambda \approx \rho - \frac{a_N}{N} + \gamma_N c N^{-2/3} \chi.$$

Standard errors of γ_N and a_N come from ten batches in generation order. The Kolmogorov-Smirnov distance of $(s - \alpha_N)/\gamma_N$ to the conjectured law, the distance after the affine fit, then measures the shape alone, and is unaffected by the far tails of section 5, which move the sample moments but not the quantiles.

2.4. The Dataset and the Pipeline. The dataset consists of 1.4×10^8 samples across ten series, each sample the largest new eigenvalue of one independently generated graph or cover, all with base degree $d = 4$.

family	sizes	samples	model
simple 4-regular	$100 \leq V \leq 20,000$ (8)	5,000,000	two permutations, repaired
simple 4-regular (legacy)	$5 \times 10^4 \leq V \leq 5 \times 10^5$ (4)	100,000	two permutations, rejection
permutation multigraph	$5 \times 10^4 \leq V \leq 5 \times 10^5$ (4)	100,000	loops and multiple edges kept
$\mathbb{Z}_k$ covers, $k = 3, 4, 5$	$100 \leq n \leq 5000$ (6 each)	1,000,000	random simple 4-regular base
quaternion covers	$100 \leq n \leq 5000$ (6)	1,000,000	random simple 4-regular base
covers of K_5	$100 \leq V \leq 10,000$ (7)	5,000,000	fixed base K_5
covers of $K_5 - e$	$100 \leq V \leq 10,000$ (7)	5,000,000	fixed irregular base
covers of $K_4 - e$	$100 \leq V \leq 10,000$ (7)	1,000,000	fixed base of cycle rank two

The simple 4-regular series draws two independent uniform permutations π_1, π_2 of $\{1, \dots, V\}$, forms the graph with adjacency matrix $P_1 + P_1^T + P_2 + P_2^T$, and repairs it locally, drawing each permutation in turn and repeatedly permuting its entries at the positions that create a loop or a multiple edge, together with one random further position, until none remains, and it discards the rare disconnected samples. The legacy simple series instead imposes simplicity by rejection, redrawing the whole graph. Both produce simple 4-regular graphs that differ from the permutation model (MNS's family $G_{V,4}$, analyzed first by Friedman [35], and a random degree V cover of the bouquet of two loops) in a bounded number of positions, and the permutation model conditioned on simplicity is contiguous with the uniform simple 4-regular model [48, 105], to which Theorem 1 applies. Contiguity alone does not transfer a limit law, and it is the data of section 3 that shows both series following the law of Theorem 1 with its constant.

The permutation multigraph series keeps the loops and multiple edges, with a loop contributing 2 to the diagonal. The abelian series draw a fresh random simple 4-regular base graph on n vertices for each batch, assign each base edge an independent uniform voltage in $\mathbb{Z}_k$, and record the largest new eigenvalue of the explicit kn -vertex cover, ten covers per base graph. A one way analysis of variance across the ten covers sharing a base gives intraclass correlations below 3×10^{-4} in absolute value at the largest size, and below 2.4×10^{-3} at every size, and so the residual randomness of the base is invisible at the fluctuation scale and the samples may be treated as independent. The quaternion series assigns each edge of a random simple 4-regular base an independent uniform element of Q_8 and records the top eigenvalue of the resulting $4n$ dimensional real block. The fixed base series assign each edge of K_5 , of K_5 minus an edge, and of K_4 minus an edge an independent uniform permutation in S_k , and record the largest new eigenvalue of the cover. These families are analyzed in section 5.

We compute eigenvalues with the implicitly restarted Lanczos method (ARPACK), applied to the adjacency matrix shifted by a small multiple of the identity so that the algebraically largest eigenvalues dominate in magnitude. We remove old eigenvalues by comparing against the base spectrum, computed exactly, with a tolerance of 10^{-10} . The number of computed eigenvalues per graph starts at 2 and grows automatically whenever every computed eigenvalue is old. For the regular families this removes the trivial eigenvalue d and yields λ_2 . For a disconnected cover, whose spectrum contains that of B twice, removal by value discards the second copy as well, an event of probability $O(k^{1-r(B)})$ by Proposition 5 below, where $r(B)$ is the cycle rank of B . Two precision caveats apply. The legacy series were generated years earlier in MATLAB and are recorded on a 10^{-12} grid, coarser than machine precision. A per process seeding scheme caused seed collisions between worker tasks in the abelian series: of its 1.8×10^6 base graph batches, 452,508 (25%) are redundant copies of another batch, concentrated at the smallest sizes (55 to 81% of batches at $n \in \{100, 200\}$, and below 0.1% at $n \geq 1000$), so that the effective number of independent samples at the smallest sizes is smaller than the nominal 10^6 . Duplicated batches agree to better than 3×10^{-13} , and dropping them moves every statistic quoted in this paper by at most two units in its final displayed digit, consistent with the sampling fluctuation of the removed samples. We verified every summary statistic by independent recomputation from the raw samples. The complete dataset, the generation code, and the analysis pipeline that produces every figure and table of this paper are available at <https://github.com/enaslund/random-graphs>.

3. THE REGULAR GRAPH CONTROL EXPERIMENT

In this section we test the pipeline against Theorem 1 along the two simple 4-regular series, measure the finite size law, diagnose the artifact behind MNS's 52%, and show that the permutation multigraph, whose moments misbehave, has the same law in its bulk.

3.1. Convergence to the Proven Law. For $d = 4$ we have that $c(4) = 3^{-1/6} = 0.8326831\dots$, and so the proven amplitudes of the mean gap and standard deviation are

$$(3.1) \quad 2\sqrt{3} - \mu_V \approx 1.0046602 V^{-2/3}, \quad \sigma_V \approx 1.0558282 V^{-2/3}.$$

The most direct test uses no fitted parameter at all. Figure 3.1 compares the distribution function of $s = (\lambda_2 - 2\sqrt{3})V^{2/3}/c(4)$ with F_1 at five sizes: the scale is right, with the

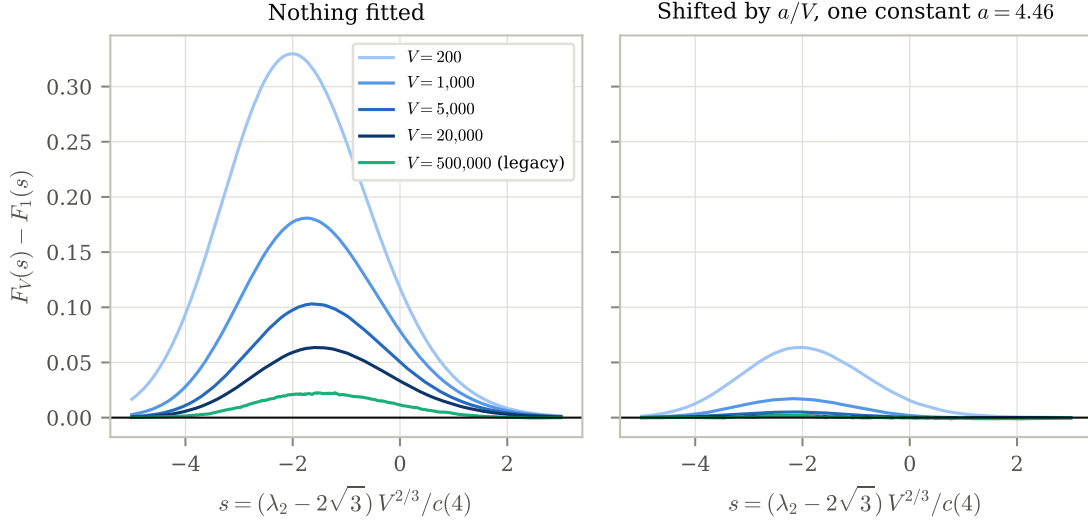


FIGURE 3.1. The random simple 4-regular graphs against Theorem 1 with nothing fitted: $F_V(s) - F_1(s)$, where F_V is the empirical distribution function of $s = (\lambda_2 - 2\sqrt{3})V^{2/3}/c(4)$. Left: the raw differences, a shift which shrinks with V . Right: the same after shifting every size by a/V , with the single constant a of subsection 3.2.

scale ratio γ_V descending from 1.117 at $V = 100$ to 1.0031 ± 0.0003 at $V = 20,000$ and equal to 1.0036 ± 0.0033 at $V = 500,000$, and a fit $c + bV^{-2/3}$ to the fitted amplitudes $\gamma_V c(4)$ extrapolates to 0.8318 to 0.8321, against $c(4) = 0.8326832$. What remains is a shift, which we analyze in subsection 3.2, and after it the differences left by the scale and shape corrections are below 0.006 from $V = 5000$ on.

The standardized statistics tell the same story. Figure 3.2 shows the standardized empirical density of 5×10^6 samples at $V = 20,000$ against the four standardized candidate laws, and Figure 3.3 the quantile-quantile plot of the largest legacy dataset, $V = 500,000$, against $\widetilde{TW}_1$ across the probability range 10^{-4} to 0.999. The Kolmogorov-Smirnov distances in Figure 3.4 quantify the picture across all sizes: the distance to $\widetilde{TW}_1$ falls with V to the sampling floor, reaching 8.7×10^{-4} at $V = 20,000$ (the floor at 5×10^6 samples is roughly 4×10^{-4}), while the distances to $\widetilde{TW}_2$, $\widetilde{TW}_4$, and the normal stall at 3.9×10^{-3} , 7.8×10^{-3} , and 1.9×10^{-2} respectively. At the legacy sizes the sample is statistically indistinguishable from the law: at $V = 500,000$ the Kolmogorov-Smirnov test accepts $\widetilde{TW}_1$ with $p = 0.99$. The mass to the left of the mean, MNS's decisive statistic, reaches 0.51885 at $V = 20,000$ and 0.52011 at $V = 500,000$ against the $\widetilde{TW}_1$ value 0.519652 from (2.2), with the competing values 0.515016, 0.511069, and 0.5 excluded by more than fifteen binomial standard errors (see Figure 4.4 in section 4). The sample skewness and excess kurtosis at the largest sizes, 0.286 and 0.156 at $V = 20,000$, converge to the $\widetilde{TW}_1$ values 0.293 and 0.165 (Figure 4.5, also in section 4).

Figure 3.5 shows the mean gap and standard deviation against V on logarithmic axes with the $V^{-2/3}$ guide, and Figure 3.6 shows the compensated quantities $(2\sqrt{3} - \mu_V)V^{2/3}$ and $\sigma_V V^{2/3}$, which approach the two constants of (3.1): at $V = 500,000$ the measured standard deviation is 1.685×10^{-4} against the predicted 1.676×10^{-4} , and the measured gap is 1.684×10^{-4} against the predicted 1.595×10^{-4} , the mean converging visibly more slowly than the standard deviation, by exactly the shift a/V of subsection 3.2. The

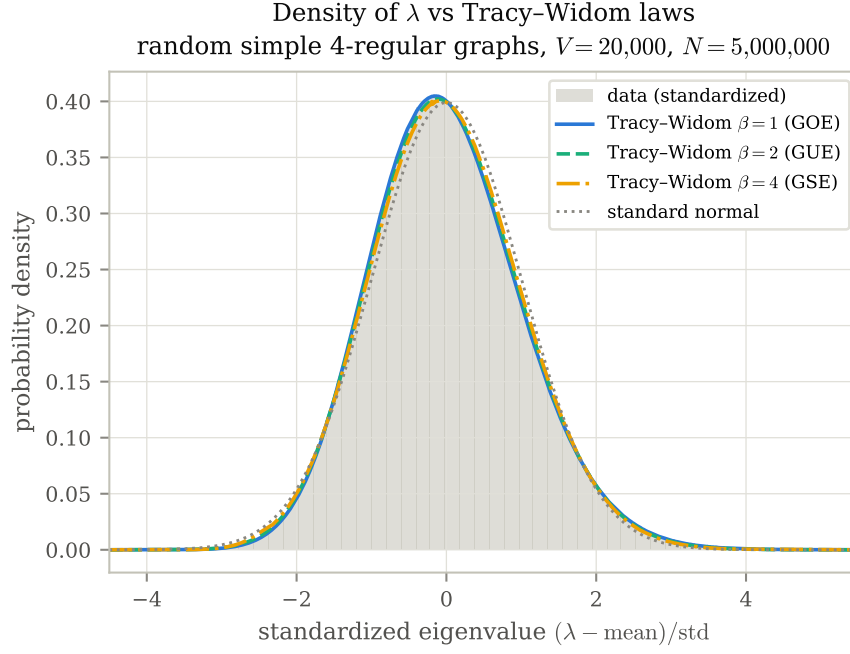


FIGURE 3.2. Standardized empirical density of the largest non-trivial eigenvalue over 5×10^6 random simple 4-regular graphs on $V = 20,000$ vertices, against the standardized Tracy-Widom $\beta = 1, 2, 4$ densities and the standard normal.

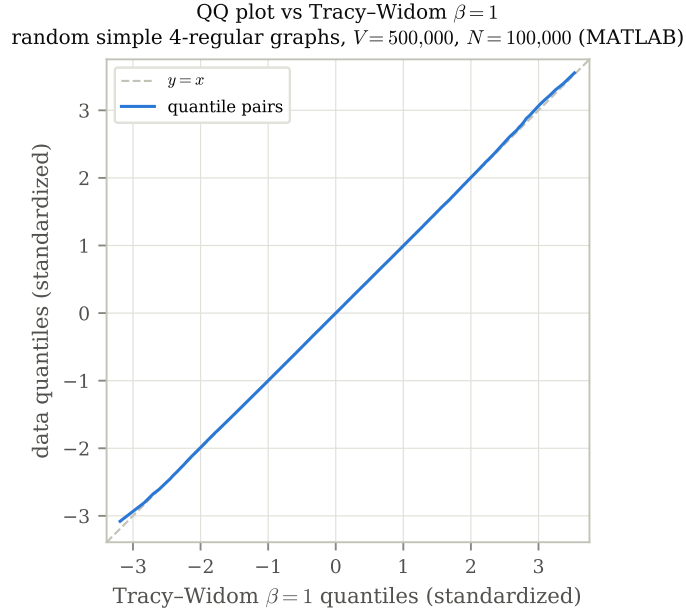


FIGURE 3.3. Quantile-quantile plot of 10^5 samples at $V = 500,000$ against $\widetilde{TW}_1$, probability range 10^{-4} to 0.999.

constants (3.1) appear in Sodin's work [98] on a closely related model in which the matrix entries carry independent random signs. We note that the arXiv posting of [98]

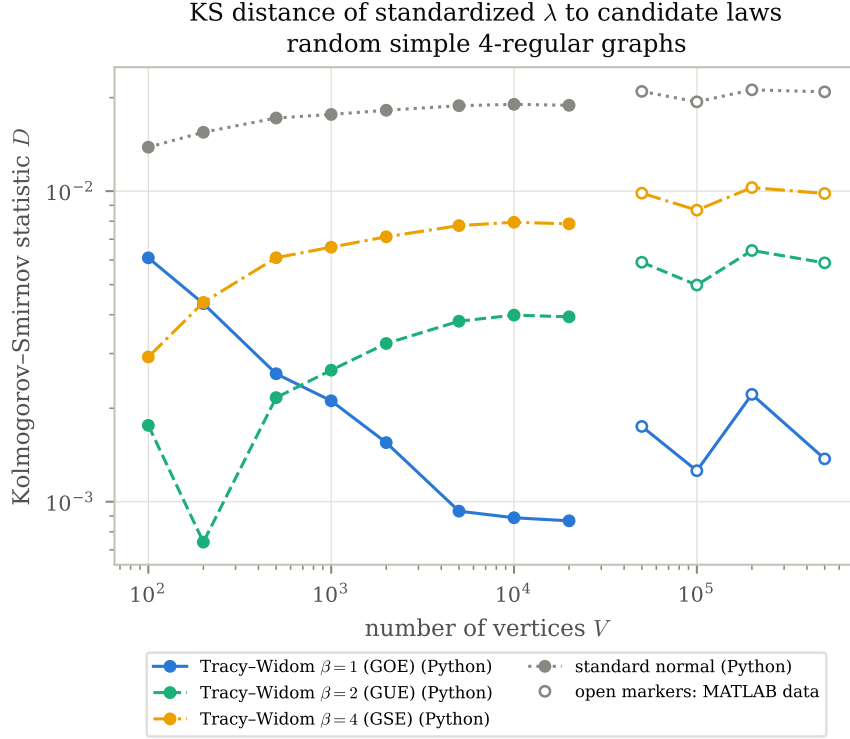


FIGURE 3.4. Kolmogorov-Smirnov distance of the standardized samples to the standardized Tracy-Widom $\beta = 1, 2, 4$ laws and the standard normal, for random simple 4-regular graphs (filled markers: 5×10^6 samples per size, open markers: legacy data with 10^5 samples). Only the distance to $\beta = 1$ keeps falling.

was withdrawn over an error in one of its lemmas, and the published version carries a 2017 erratum, so that Theorem 1 is the rigorous source for (1.2).

3.2. The Finite Size Law. The theorems give no rate of convergence, and the data is precise enough to measure the finite size law itself. Figure 3.1 shows that its leading term is a shift: with a_V the location coefficient of (2.6), the simple 4-regular series gives $a_V = 5.24$ at $V = 100$, 4.54 at $V = 1000$, and then 4.479 , 4.461 , and 4.453 at $V = 5000$, $10,000$, and $20,000$, with standard errors between 0.01 and 0.02 , and so, with a the weighted mean over $V \geq 5000$,

$$(3.2) \quad \lambda_2 \approx 2\sqrt{3} - \frac{a}{V} + c(4)V^{-2/3}TW_1, \quad a = 4.465 \pm 0.008.$$

In units of the fluctuation scale this is a shift of $a/(c(4)V^{1/3})$, which is 0.20 at $V = 20,000$ and 0.07 at $V = 500,000$, and the scale correction is much smaller, with $\gamma_V - 1$ decaying roughly like $V^{-2/3}$. The shift is not a property of the degree alone. The permutation multigraph, the same model without the conditioning, has $a = -1.44 \pm 0.10$ (subsection 3.4), the covers of the 4-regular graph K_5 have a between 0.01 and 0.07 from $V = 1000$ on, and the legacy simple series, which imposes simplicity by rejection rather than by local repair, gives $a = 4.87 \pm 0.12$ over its four sizes, a difference of 3.4 standard errors from (3.2) that we do not regard as established. Conditioning on simplicity thus moves the whole law down by about $6/V$. A shift of order $1/V$ is what one expects from the short cycles: in the physics literature, Metz, Parisi, and Leuzzi [82] computed the

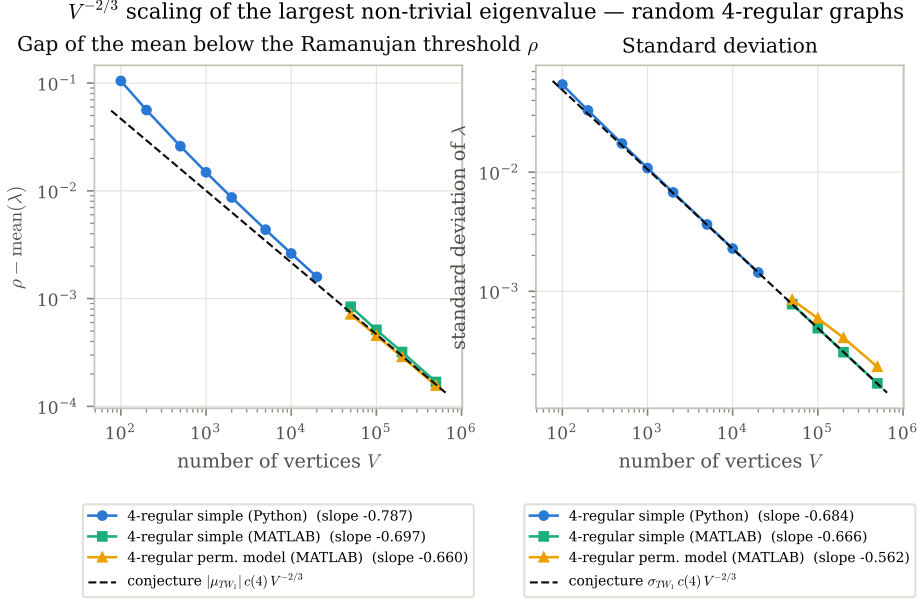


FIGURE 3.5. Mean gap $2\sqrt{3} - \mu_V$ and standard deviation against V for the three 4-regular series, with fitted slopes and the $V^{-2/3}$ Tracy-Widom line.

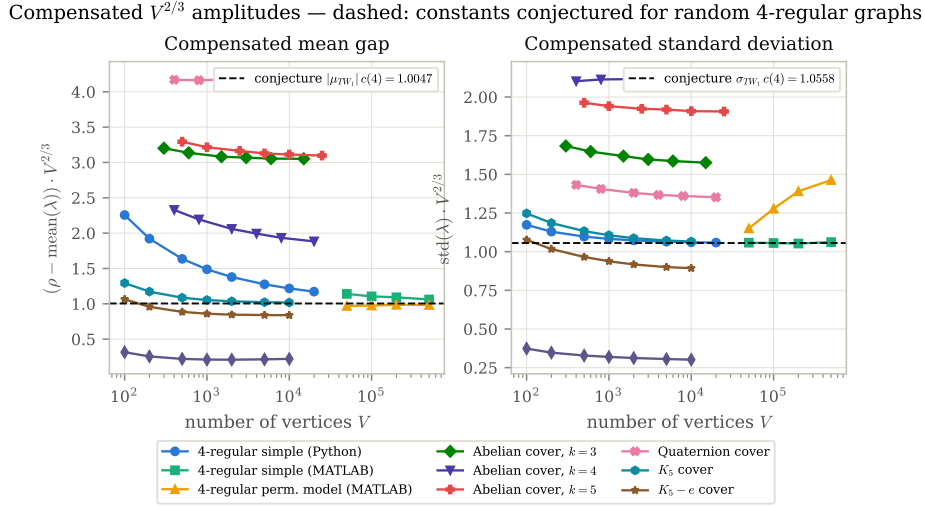


FIGURE 3.6. Compensated scaling: $(\rho - \mu_V) V^{2/3}$ and $\sigma_V V^{2/3}$ against V for all families, where ρ is the family's Ramanujan threshold. The dashed lines are the regular graph amplitudes (3.1), which the simple 4-regular and K_5 cover series approach, while each remaining family plateaus at its own amplitude, predicted by Conjectures 2 and 4 (see sections 4 and 5).

$1/V$ correction to the Kesten-McKay density of random regular graphs as a sum over cycle lengths, with an inverse square root singularity at the edge, which is the form of a deterministic shift of the edge at order $1/V$, and conditioning on simplicity removes the cycles of length one and two. For sparse Erdős-Rényi graphs the corresponding deterministic shift of the edge is proven [69, 25].

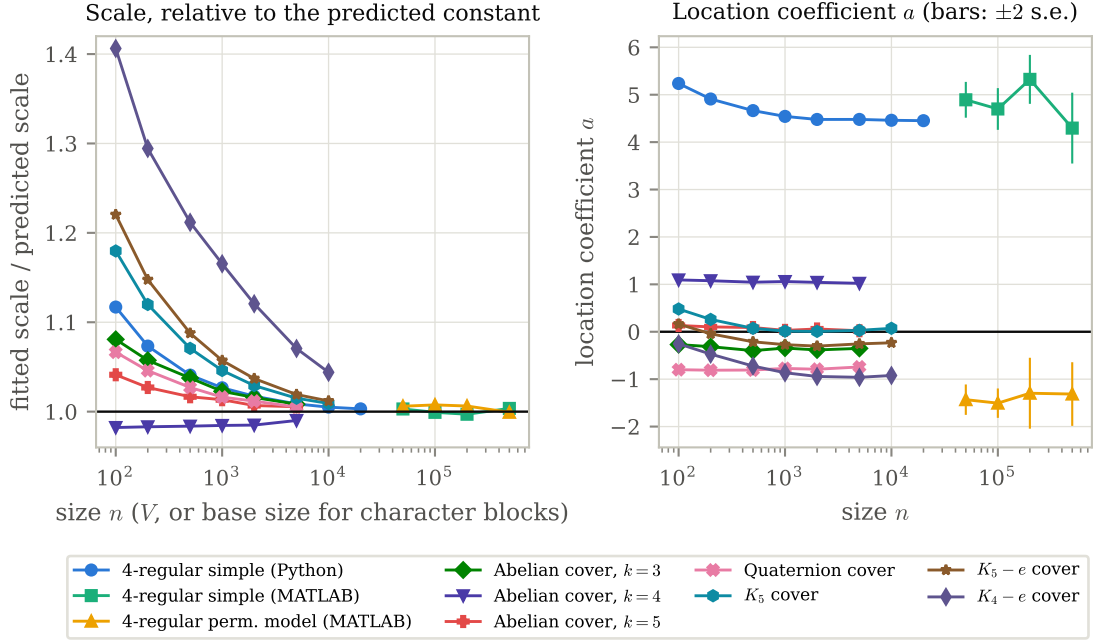


FIGURE 3.7. The two finite size parameters of (2.6) for every family, from the quantile regression of the parameter-free variable on the conjectured law. Left: the scale ratio, the fitted scale over the predicted one, which is $c(4)$ for the 4-regular families, $c(4)2^{-1/6}$ in terms of TW_4 for the quaternion covers, and $C_B^{-2/3}$ of (2.4) for $K_5 - e$ and $K_4 - e$. Right: the location coefficient a , in the law $\rho - a/N + \gamma c N^{-2/3} \chi$, which settles at a family dependent constant.

The shift controls the Ramanujan probability, which by (3.2) is

$$(3.3) \quad P\left(\lambda_2 \leq 2\sqrt{3}\right) \approx F_1\left(\frac{a}{c(4)V^{1/3}}\right) = F_1(0) + \frac{0.9728}{V^{1/3}} + O(V^{-2/3}),$$

where $0.9728 = f_1(0)a/c(4)$ and $f_1(0) = 0.18142$ is the density of TW_1 at the origin. With the single constant (3.2), the left side of (3.3) matches the data to within 3×10^{-4} , or 2.4 binomial standard errors, at every size from $V = 1000$ to $20,000$, and to within 2.6 standard errors at the four legacy sizes. The convergence to $F_1(0)$ is thus slow: the Ramanujan probability of the simple model is still 0.01 above its limit at $V = 8.6 \times 10^5$, and reaches within 10^{-3} only at $V = 9 \times 10^8$ (Figure 3.8). For the multigraph the approach is from below, since $a < 0$, and it is within 10^{-3} at $V = 3 \times 10^7$, while the covers of K_5 , with a nearly zero, are within 10^{-3} from $V = 1000$ on. Any experiment on the Ramanujan probability at accessible sizes measures F_β at a shifted point, and the shift must be accounted for.

The third correction is to the shape. Rates at the Tracy-Widom edge have a history in random matrix theory: Choup [30] computed the Edgeworth expansion of the largest eigenvalue law of the finite GOE, Johnstone and Ma [65] proved that the rate of approach depends on the centering convention, Bornemann [22] expanded the laws of the finite Gaussian ensembles at the soft edge in terms of the derivatives of F_β , and rates of order $N^{-1/3}$ are proven for Wigner matrices [97] and sparse Erdős-Rényi graphs [25]. Write F_V for the distribution function of the standardized largest non-trivial eigenvalue

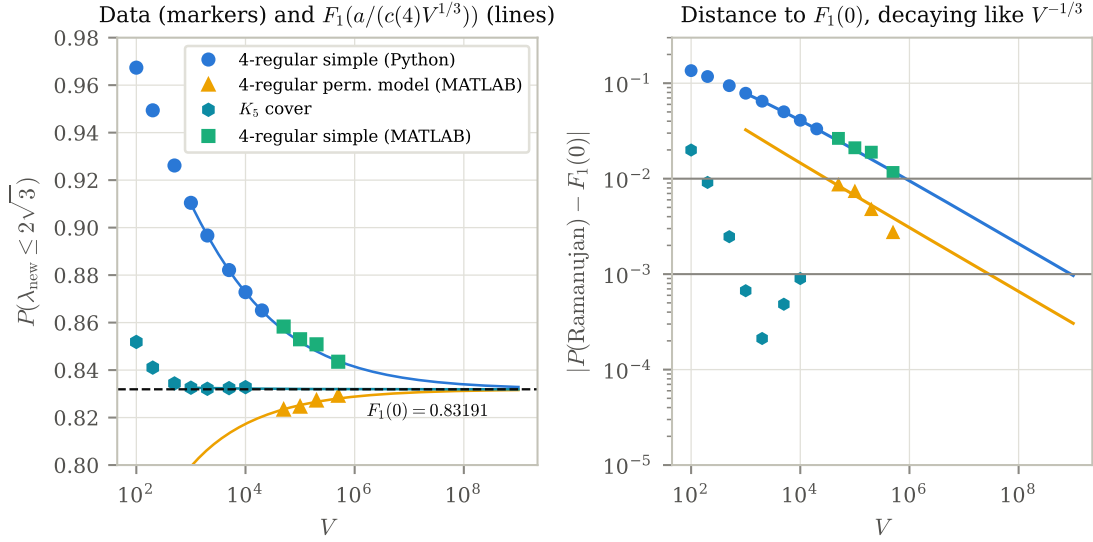


FIGURE 3.8. Left: the empirical Ramanujan probability against V for the simple, legacy simple, multigraph, and K_5 cover series (markers), with the one parameter model $F_1(a/(c(4)V^{1/3}))$ of (3.3) (lines, drawn from $V = 1000$). Right: the distance to $F_1(0)$ on logarithmic axes, extrapolated to $V = 10^9$, with the levels 10^{-2} and 10^{-3} .

and $\tilde{F}_1, \tilde{f}_1$ for the standardized Tracy-Widom distribution and density. Standardization removes the location and scale, and across the simple 4-regular family the remaining deviation $F_V - \tilde{F}_1$ is, to within 6 to 14 percent of its squared norm, proportional to the single fixed profile $(t^2 - 1)\tilde{f}_1(t)$, a correction of Edgeworth type signaling a skewness deficit, and so to the resolution of 5×10^6 samples the finite size law is the one parameter family

$$F_V(t) \approx \tilde{F}_1(t) - \hat{c}(V) (t^2 - 1) \tilde{f}_1(t),$$

where the fitted amplitude $\hat{c}(V)$ decays like a power law with exponent -0.46 ± 0.04 , and the K_5 cover family of section 5 lies on the same curve (Figure 3.9). The same parameter accounts for the residuals of the individual standardized statistics: the mass left of the mean at $V = 20,000$ still sits -8.0×10^{-4} from the TW_1 value, 3.6 binomial standard errors, against an Edgeworth prediction of $-\hat{c}\tilde{f}_1(0) = -5.3 \times 10^{-4}$, and after subtracting the fitted profile the left tails of the largest datasets agree with $\tilde{TW}_1$ to within three standard errors down to probability 6×10^{-7} .

3.3. The Origin of the 52% Conjecture. MNS reached the first case of (2.5) by fitting single power laws to the mean gap and standard deviation over their full range of sizes. Our data reproduces their fit and (3.2) explains it. Fitting one line to $\log(2\sqrt{3} - \mu_V)$ against $\log V$ over our eight sizes $V \leq 20,000$, the MNS range, gives a slope of -0.787 , while the same fit to $\log \sigma_V$ gives -0.684 : a mean exponent apparently steeper than the standard deviation exponent, exactly the pattern from which MNS concluded $g_V \rightarrow 0$ and hence 52%, with numbers closely matching their $d = 3$ values $m \approx -0.79$ and $s \approx -0.72$. Under (3.2) the mean gap is $AV^{-2/3} + a/V$ with

The deviation of the standardized ECDF from TW_1 is a single skewness profile with power-law amplitude

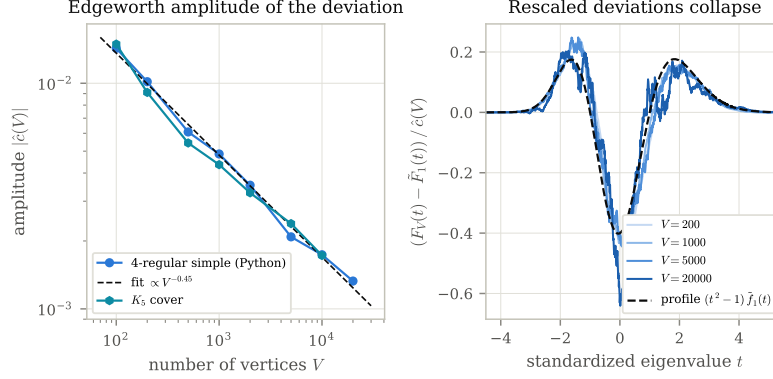


FIGURE 3.9. The shape correction. Left: the amplitude $\hat{c}(V)$ of the deviation of the standardized empirical distribution from $\widehat{TW}_1$, for the simple 4-regular and K_5 cover families, with the fitted power law. Right: the deviations at four sizes, rescaled by $\hat{c}(V)$, collapse onto the fixed profile $(t^2 - 1)\tilde{f}_1(t)$ of a skewness deficit.

$A = 1.0046602$ from (3.1), whose local exponent is

$$\frac{d \log (2\sqrt{3} - \mu_V)}{d \log V} = -\frac{2/3 + x}{1 + x}, \quad x = \frac{a}{AV^{1/3}},$$

which is -0.72 at $V = 14,000$ and approaches $-2/3$ from below only as $V^{-1/3}$. Figure 3.10 compares this with the slopes between consecutive sizes. The standard deviation exponent sits within 0.01 of $-2/3$ from $V \approx 2000$ onward, while the mean gap exponent drifts monotonically from -0.90 at $V \approx 100$ to -0.72 at $V = 20,000$ within the five million sample series, with the legacy series continuing the descent and reaching -0.70 over its last pair at $V = 500,000$, still short of $-2/3$, all along the curve of the one constant model. Fitted to the model at our eight sizes, a single power law gives -0.764 , most of the observed -0.787 , the remainder coming from the larger values of a_V at the smallest sizes. MNS saw the warning sign, observing that their best fit exponents increased monotonically as sizes were added, and were appropriately cautious. At $V \leq 20,000$ the data could do no better: by (3.3) the Ramanujan probability itself is 0.865 at $V = 20,000$, and no single size in that range is close to the limit. The ratio g_V resolves the question directly: as Figure 4.12 shows, g_V does not tend to 0 for any family, and for the simple 4-regular series it descends to 0.9993 at $V = 500,000$, approaching the Tracy-Widom ratio $|\mu_{TW_1}|/\sigma_{TW_1} = 0.951537\dots$ from above, in agreement with the middle case of (2.5) and with $F_1(0) = 0.8319081\dots$ as the limiting Ramanujan probability.

3.4. The Multigraph Model. MNS pooled models with and without loops and multiple edges, observing similar behavior at their sample sizes. At ours, the sample moments of the permutation multigraph series, identical to the legacy simple series except that loops and multiple edges are kept, differ qualitatively: at $V = 500,000$ its skewness is 74.5 and its excess kurtosis 1.15×10^4 , against the TW_1 values 0.293 and 0.165, its ratio g_V falls through 0.67, and the Kolmogorov-Smirnov distance of the standardized sample to every candidate law grows with V , reaching 0.085. None of this comes from the bulk. With nothing fitted, the parameter-free variable s of the multigraph is closer

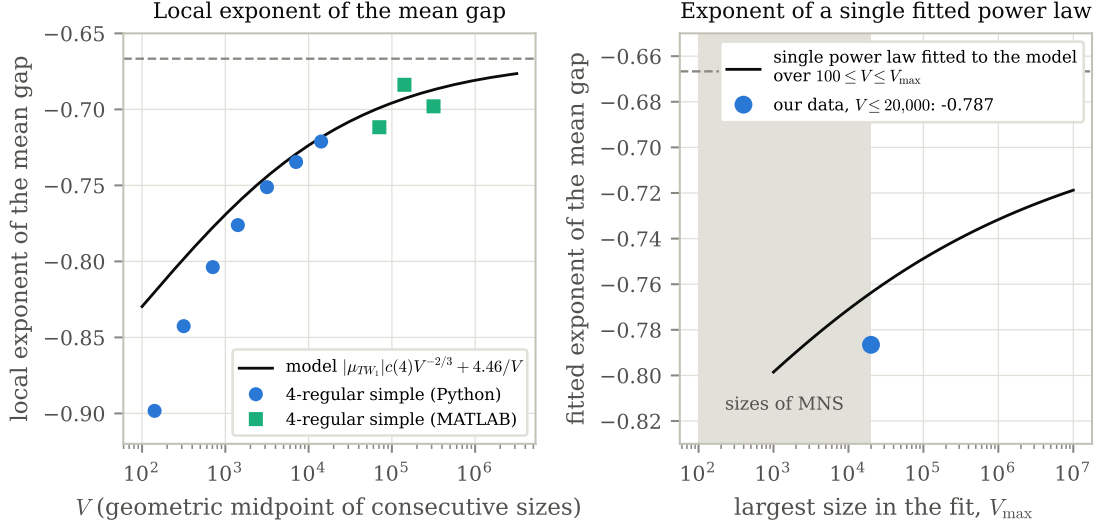


FIGURE 3.10. The origin of the 52% conjecture. Left: the local exponents of the mean gap between consecutive sizes for the two simple 4-regular series, with the prediction of the one constant model (3.2) and the limit $-2/3$ dashed. Right: the exponent reported by a single power law fitted to the model over $100 \leq V \leq V_{\max}$, against $V_{\max}$, with the value -0.787 fitted to our data over the sizes of MNS (shaded).

Local log-log slopes between consecutive sizes — a single power-law fit over small V overestimates the gap exponent

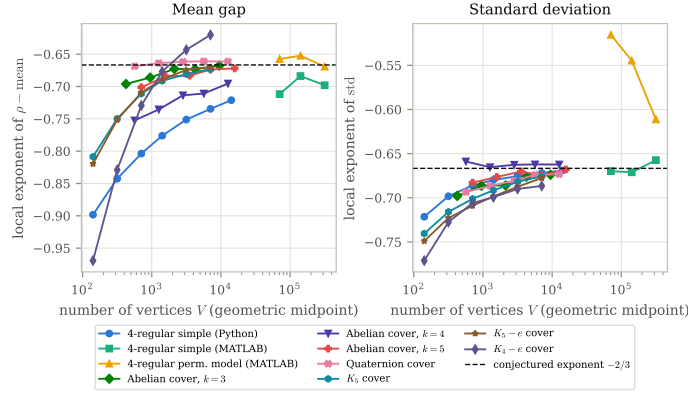


FIGURE 3.11. Local (consecutive size) log-log exponents of the mean gap and the standard deviation for all series, with the $-2/3$ line. The mean gap exponent approaches $-2/3$ from below as V grows, and so a single power law fitted over small sizes overestimates its steepness, wrongly suggesting that the gap outruns the standard deviation.

to TW_1 than that of the simple model (Figure 3.12, left), at Kolmogorov-Smirnov distance 0.0077 against 0.0227 at $V = 500,000$, since its location coefficient is smaller, $a = -1.44 \pm 0.10$ over the four sizes, and its scale ratio is 0.9992 ± 0.0027 . After the affine fit of subsection 2.3 the distance is 0.0016, below the typical value 0.0027 for 10^5 samples drawn from the law itself. The moments are instead carried by a handful of far outliers, a few samples in each 10^5 , up to about 200 standard deviations of the bulk

above its mean at $V = 500,000$, which inflate the sample standard deviation by about 40% and so compress the standardized bulk.

The outliers are localized eigenvalues, and they are not produced by the loops and multiple edges alone. A loop contributes 2 to the diagonal, and a loop, or a double edge, attached to the branches of the 4-regular tree, binds no eigenvalue above $2\sqrt{3}$, since the resulting equation $\lambda = 2 + 2y(\lambda)$, with $y(\lambda) = (\lambda - \sqrt{\lambda^2 - 12})/6$ the branch value of (2.3), has no solution above $2\sqrt{3}$, where its right hand side is $2 + 2/\sqrt{3} = 3.155$. In the language of subsection 5.2, the binding defects have cycle rank two. A triple edge, and two looped vertices joined by an edge, each bind at exactly $7/2$, a looped vertex joined to its neighbor by a double edge binds at the real root

$$\lambda = 3.5385847 \dots \quad \text{of} \quad \lambda^3 = 8\lambda + 16,$$

and a triangle with a loop at one vertex binds just above the threshold, at $3.4675039 \dots$, and an exhaustive search shows that no other defect of cycle rank two on at most five vertices binds. Of the 18 samples more than eight standard deviations above $2\sqrt{3}$ across the four legacy sizes, 17 lie within 0.002 of these three values (Figure 3.12, right), with those on $7/2$ draining up to it from 3.4985 at $V = 50,000$ to 3.4999 at $V = 500,000$, and the exception, 3.5116 at $V = 50,000$, lies among the values of several defects of cycle rank three. Defects of cycle rank two occur at rate $1/V$ by Proposition 5, and so the multigraph model has least binding rank $r^* = 2$, as does $K_4 - e$, and Corollary 8 applies: the defects eventually dominate the variance, growing relative to the bulk like $V^{1/3}$, which is the growth of every moment based distance observed above. Conditioning on simplicity removes all three defects, and no simple subgraph of cycle rank two on at most five vertices binds, so that the leading defect of the simple model is K_4 , which binds at $7/2$ as in the covers of K_5 but has cycle rank three and occurs at rate V^{-2} . The largest sample of the simple series at $V = 500,000$ sits only 9 standard deviations above its mean.

The lesson for experiments in this area is that statistics built from sample moments are unusable when defects of low cycle rank bind, while quantile statistics are immune to them. The simple and multigraph models share the limiting spectral distribution, the almost sure convergence of λ_2 to $2\sqrt{3}$, and the Tracy-Widom law of the bulk with its constant, and they are separated at the fluctuation scale by a deterministic shift of order $1/V$ and by a thin tail of defects of cycle rank two.

4. COVERS AND THE THREEFOLD WAY

We now turn to the main experiment. Each abelian and quaternion family isolates a definite set of isotypic blocks, and Conjecture 2 predicts the symmetry class of each: $\mathbb{Z}_3$ covers should follow χ_2 , quaternion covers χ_4 , the regular graph families of section 3 χ_1 , and the mixed families $\mathbb{Z}_4$ and $\mathbb{Z}_5$ no single Tracy-Widom law at all.

4.1. Ensemble Selection in the Pure Families. Figure 4.1 is the structural result of this paper. For each family at its largest size, and for each candidate law, it shows the Kolmogorov-Smirnov distance after the best affine fit to that law, which removes the location and scale and compares the shapes alone. In every row the minimum is the law of the family's own class: TW_1 for the regular, K_5 , and $K_5 - e$ families, at 0.6 to 0.7×10^{-3} , against 2.1×10^{-3} or more for every other law, TW_2 for $\mathbb{Z}_3$ covers, at 0.8×10^{-3} against 2.6×10^{-3} for TW_4 and 3.5×10^{-3} for TW_1 , and TW_4 for quaternion covers, at 0.5×10^{-3} against 2.7×10^{-3} for TW_2 and 5.6×10^{-3} for TW_1 , where the

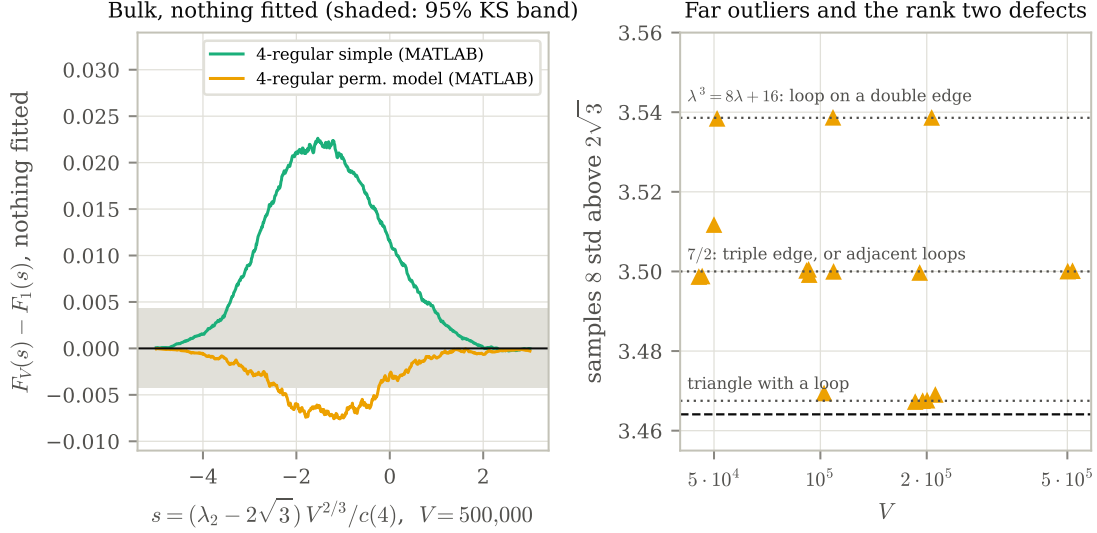


FIGURE 3.12. The permutation multigraph model. Left: $F_V(s) - F_1(s)$ for the parameter-free variable at $V = 500,000$, for the multigraph and the legacy simple series, with the 95% Kolmogorov-Smirnov band for 10^5 samples shaded: nothing is fitted, and the multigraph is closer to the law. Right: every multigraph sample more than eight standard deviations above $2\sqrt{3}$ against V , with the three computed defect eigenvalues of cycle rank two dotted.

typical distance of 10^6 samples from their own law is 0.9×10^{-3} . The mixed families select their maximum laws in the same way (subsection 4.3).

The standardized statistics agree. Figure 4.3 shows that for each pure family the Kolmogorov-Smirnov distance to the standardized law of its own symmetry class falls to the sampling floor while the distance to $\widetilde{TW}_1$ (dotted) stalls. At the largest sizes, the $\mathbb{Z}_3$ family reaches distance 1.0×10^{-3} from $\widetilde{TW}_2$ (floor at 10^6 samples: roughly 9×10^{-4}) while its distance to $\widetilde{TW}_1$ is stuck at 5.4×10^{-3} , and the quaternion family reaches 7.5×10^{-4} from $\widetilde{TW}_4$ with distance 8.8×10^{-3} to $\widetilde{TW}_1$. The mass to the left of the mean (Figure 4.4) separates the classes at higher resolution: the $\mathbb{Z}_3$ series converges to 0.5145 against the TW_2 value 0.515016, the quaternion series to 0.5109 against the TW_4 value 0.511069, and the regular series to 0.5201 against the TW_1 value 0.519652, each matching its own class, with every competing class excluded by six or more binomial standard errors for the cover families and by three or more for the legacy regular sample.

The tails separate the classes most visibly, since the standardized Tracy-Widom laws differ most there, the left tail of TW_β decaying like $\exp(-\beta|s|^3/24)$. Figure 4.2 shows both tails of the three pure families down to probability 10^{-6} , after mapping each sample onto its own law by the affine fit and standardizing: each follows the tail of its own class. The affine invariant moments agree as well: Figure 4.5 shows the sample skewness and excess kurtosis of each family converging to the values of its own class, (0.213, 0.093) at the largest $\mathbb{Z}_3$ size against the TW_2 values (0.224, 0.093), and (0.156, 0.042) at the largest quaternion size against the TW_4 values (0.166, 0.049), and Figure 4.6 shows the same trajectories in the plane of the two moments, where each family, the mixed ones included, moves to the point of its own limit law. Figure 4.7 gives the quantile level

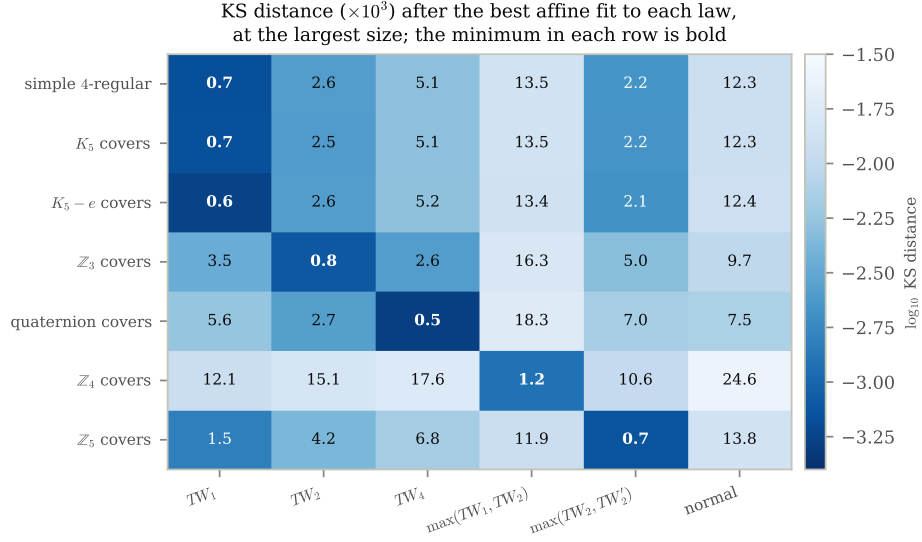


FIGURE 4.1. Kolmogorov-Smirnov distance, times 10^3 , between each family at its largest size and each candidate law, after the best affine fit of the sample to that law, so that only the shape is compared. The minimum in each row, in bold, is the law predicted by Conjectures 2, 3, and 4. The typical distance of a sample from its own law is 0.4×10^{-3} at 5×10^6 samples and 0.9×10^{-3} at 10^6 .

view of the same selection, and Figure 4.8 overlays the standardized densities of the largest datasets from five families on the $\widehat{TW}_1$ density: at the resolution of the eye all collapse to the Tracy-Widom shape, and the discrimination between the β classes lives in the statistics above.

4.2. The Scale Constants. Conjecture 2 predicts the scale as well as the class, and the parameter-free variable of subsection 2.3 tests it with the constant $c(4)$ of random 4-regular graphs and the base size n as the block dimension. For the character blocks this rests on the fact that each block A_χ is the adjacency matrix of the random base graph with its entries rotated by the unimodular phases $\chi(\sigma_e)$, a perturbation which does not change the tree-like local structure or the Kesten-McKay law. The data confirms the prediction for every family (Figure 3.7). At $n = 5000$ the scale ratios are 1.0083 ± 0.0006 for $\mathbb{Z}_3$, 1.0055 ± 0.0013 for $\mathbb{Z}_5$, and 0.9901 ± 0.0012 for $\mathbb{Z}_4$, each converging monotonically to 1, and 1.0052 ± 0.0008 for the quaternion covers, against the constant $c(4)$ with the GSE normalization $\chi_4 = 2^{-1/6}TW_4$ of subsection 2.2. Without the factor $2^{-1/6}$ the quaternion ratio would be 0.896, and so the data selects the GSE normalization of the Gaussian ensembles, and the quaternionic block carries the same constant as every other block. Figure 4.9 shows the parameter-free distribution functions against the conjectured laws: the differences shrink with n in every family, and after the affine fit the Kolmogorov-Smirnov distances at the largest size are between 0.5×10^{-3} and 1.2×10^{-3} , at the sampling floor. The location coefficients settle at family dependent constants, $a = -0.35$ for $\mathbb{Z}_3$, $+1.05$ for $\mathbb{Z}_4$, $+0.05$ for $\mathbb{Z}_5$, and -0.77 for the quaternion covers, each with standard error near 0.01, all much smaller than the 4.47 of the simple graphs.

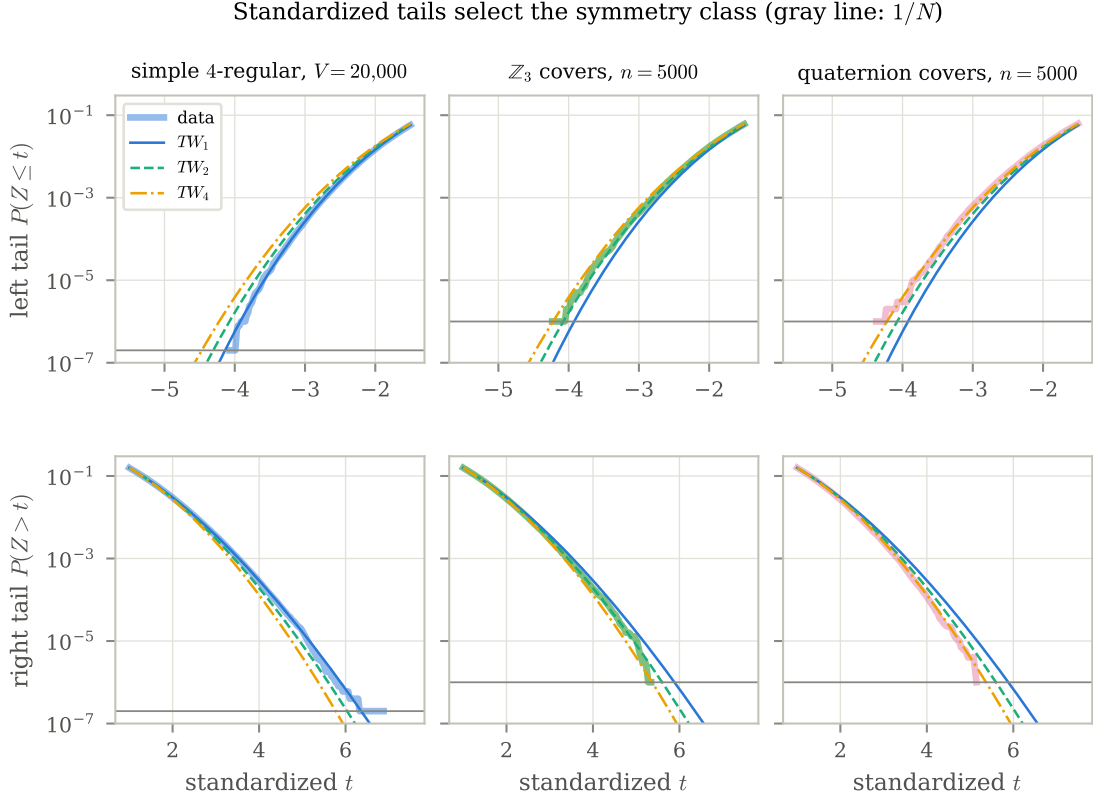


FIGURE 4.2. Left tails (top) and right tails (bottom) on logarithmic axes for the simple 4-regular graphs at $V = 20,000$, the $\mathbb{Z}_3$ covers at $n = 5000$, and the quaternion covers at $n = 5000$, after the affine fit to the law of each family's class and standardization, against the standardized TW_1 , TW_2 , and TW_4 . The gray line is $1/N$.

4.3. The Mixed Families and the Maximum Law. The largest new eigenvalue of a $\mathbb{Z}_4$ cover is the maximum of a GOE class maximum and a GUE class maximum, and that of a $\mathbb{Z}_5$ cover is the maximum of two GUE class maxima, and so under Conjectures 2 and 3 neither should converge to any single Tracy-Widom law. This is visible in the raw statistics: the Kolmogorov-Smirnov distance of the standardized $\mathbb{Z}_4$ family to every candidate law grows with V , reaching 1.9×10^{-2} against $\widetilde{TW}_1$ at the largest size, and the $\mathbb{Z}_5$ family, nearest $\widetilde{TW}_1$ at moderate sizes, drifts away from it as V grows.

The maximum structure can be tested quantitatively. If the two blocks of a family fluctuate independently with the same centering $2\sqrt{d-1}$ and the same scale constant, as Conjecture 2 asserts, then the largest new eigenvalue converges, on the scale $c(d)n^{-2/3}$, to

$$M = \max(X_1, X_2),$$

where X_1, X_2 are independent Tracy-Widom variables of the appropriate classes, with distribution function $F_1 F_2$ (for $\mathbb{Z}_4$) and F_2^2 (for $\mathbb{Z}_5$). After the affine fit, the Kolmogorov-Smirnov distance of each family to its maximum law falls with n to 1.2×10^{-3} for $\mathbb{Z}_4$ and 0.7×10^{-3} for $\mathbb{Z}_5$ at $n = 5000$, while the distances to the pure laws grow (Figure 4.10). Note that the drift away from the pure laws is invisible at moderate sizes: up to $n = 2000$ the $\mathbb{Z}_5$ family is closer in shape to TW_1 than to its own maximum law, whose skewness and kurtosis are close to those of TW_1 , and with 10^6 samples at $V = 2500$

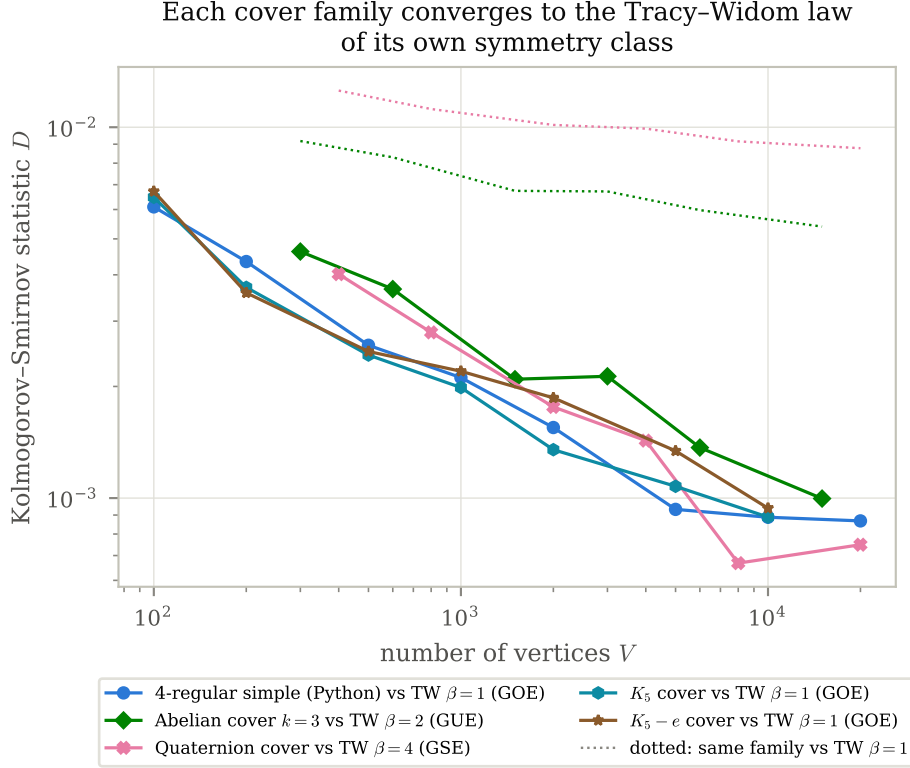


FIGURE 4.3. Kolmogorov-Smirnov distance of each pure family to the standardized Tracy-Widom law of its own symmetry class (solid): regular graphs and the fixed base K_5 and K_5-e cover families to $\beta=1$, $\mathbb{Z}_3$ covers to $\beta=2$, quaternion covers to $\beta=4$, with the distance to $\beta=1$ dotted for contrast. The matching class distances fall to the sampling noise floor. The mixed families $\mathbb{Z}_4$ and $\mathbb{Z}_5$ match no single law.

the Kolmogorov-Smirnov test accepts $\widetilde{TW}_1$ ($p = 0.81$), a caution for experiments at the scale of MNS.

Computing the distribution functions $F_1 F_2$ and F_2^2 numerically gives the predicted constants in Table 1: the ratio $|\mu_M|/\sigma_M$, the mass of M left of its mean, and the Ramanujan functional $F_{\beta_1}(0)F_{\beta_2}(0)$. The $\mathbb{Z}_5$ family, whose two blocks have identical marginal distributions, matches the mass and Ramanujan predictions to three decimal places at $n = 5000$, and the ratio to within 3×10^{-3} . The $\mathbb{Z}_4$ family is still drifting at $n = 5000$, in each statistic monotonically toward the predicted value, and the cause is its location coefficient $a = 1.05$, the largest among the covers. With it, the finite size law (2.6) predicts the $\mathbb{Z}_4$ Ramanujan probability $F_1 F_2(a/(c(4)n^{1/3})) = 0.82291$ at $n = 5000$, against the observed 0.82293. The maximum structure is visible at the moment level as well: at $n = 5000$ the $\mathbb{Z}_4$ skewness and excess kurtosis are (0.570, 0.658) against the $\max(TW_1, TW_2)$ values (0.584, 0.670), and the $\mathbb{Z}_5$ values are (0.320, 0.212) against (0.329, 0.225), in each case drifting monotonically toward the prediction (Figure 4.6).

The Ramanujan column of Table 1 is the experimental form of Conjecture 3. Each family approaches the product $\prod F_{\beta}(0)$ over its blocks: one GUE block gives $F_2(0) = 0.96937\dots$ for $\mathbb{Z}_3$, two give $F_2(0)^2 = 0.93968\dots$ for $\mathbb{Z}_5$, matched by the observed 0.94011 at $n = 5000$, the mixed $\mathbb{Z}_4$ product $F_1(0)F_2(0) = 0.80643\dots$ is approached from above through 0.82293, and the quaternion block gives $F_4(0) = 0.99857\dots$ against

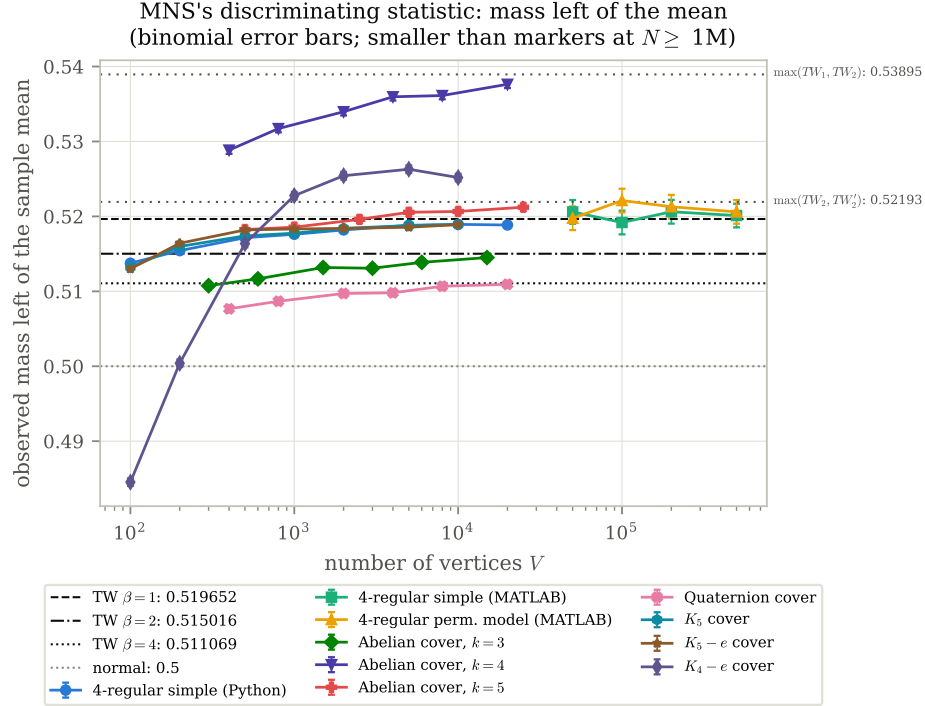


FIGURE 4.4. The fraction of samples below the sample mean against V for all families, with the Tracy-Widom $\beta = 1, 2, 4$ and normal reference values. This replicates MNS's decisive experiment at fifty times their sample size: with 10^6 to 5×10^6 samples the binomial standard error is between 2×10^{-4} and 5×10^{-4} and the four candidate laws are separated by tens of standard errors. Each pure family selects the value of its own symmetry class, while the mixed family $\mathbb{Z}_4$ drifts to ≈ 0.538 , consistent with the maximum law of Table 1 and no Tracy-Widom law.

Third and fourth moments converge to the Tracy-Widom values of each family's symmetry class (permutation model omitted: kurtosis up to $\sim 10^4$)

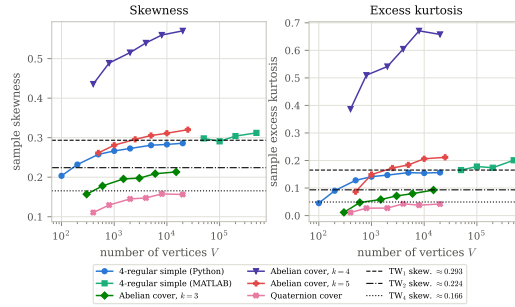


FIGURE 4.5. Sample skewness and excess kurtosis against V with the Tracy-Widom $\beta = 1, 2, 4$ reference values. The regular families converge to the $\beta = 1$ moments, $\mathbb{Z}_3$ to $\beta = 2$, and quaternion to $\beta = 4$. The multigraph model (excess kurtosis up to 10^4) and the fixed base cover families, whose third and fourth moments are dominated by the far outlier tails of section 5, are omitted.

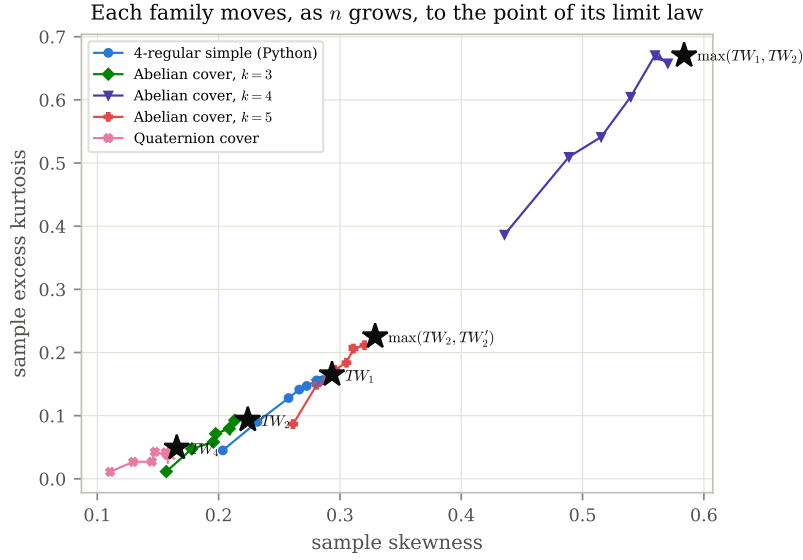


FIGURE 4.6. Sample skewness against excess kurtosis for the simple 4-regular, cyclic, and quaternion families, one marker per size, with the arrow at the largest size. Each family moves to the point of its limit law (stars): TW_1 , TW_2 , TW_4 , or the maximum laws of subsection 4.3.

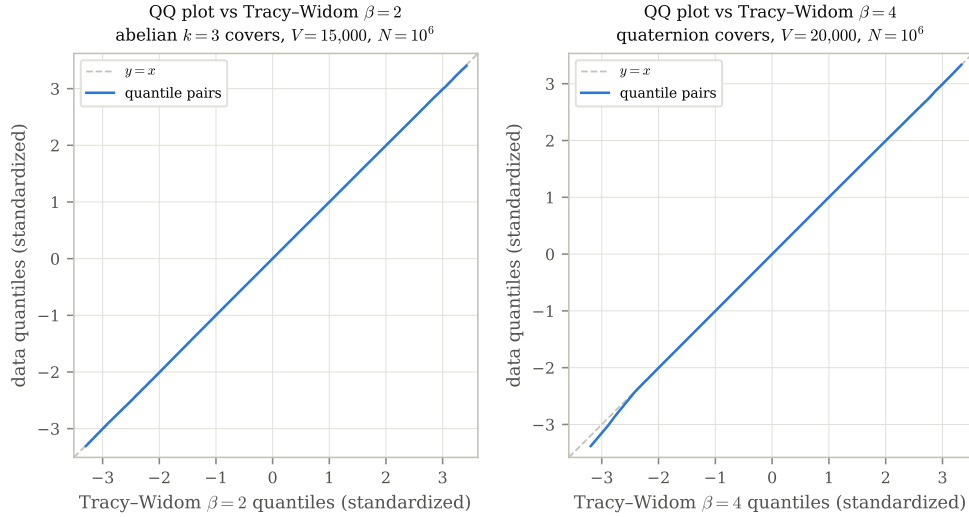


FIGURE 4.7. Quantile-quantile plots of the largest $\mathbb{Z}_3$ cover dataset against Tracy-Widom $\beta = 2$ (left) and of the largest quaternion cover dataset against $\beta = 4$ (right), both standardized, probability range 10^{-4} to 0.999.

the observed 0.99825. The residual gaps are the location shifts: evaluating each law at its shifted point $a/(cN^{1/3})$ reproduces every observed Ramanujan fraction in the table to within 3×10^{-4} , and to within 6×10^{-4} for $K_4 - e$. Figure 4.11 shows the full curves, and Figure 4.12 the underlying ratios g_V , each family flattening onto the ratio of its own limit law. In particular the limiting probability of being Ramanujan is not one number but a representation theoretic invariant of the family, always strictly between 0 and 1.

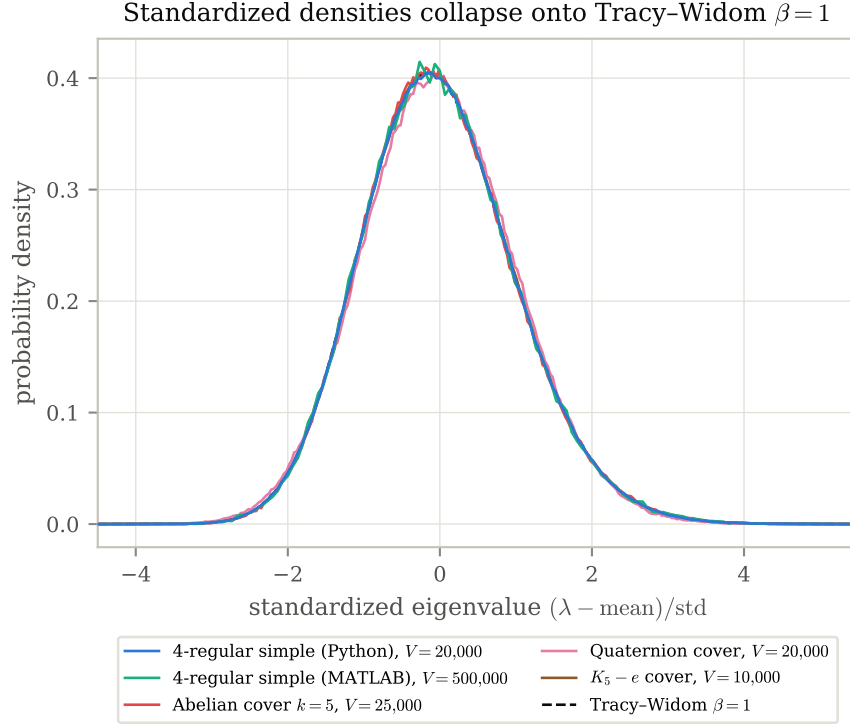


FIGURE 4.8. Standardized empirical densities of the largest datasets from five families overlaid on the standardized $\beta = 1$ Tracy-Widom density. The β classes are indistinguishable at this resolution, and are separated by the statistics of Figures 4.1, 4.2, 4.3, 4.4, and 4.5.

family	limit law	g_V	mass left of mean	$P(\text{Ramanujan})$
simple 4-regular	TW_1	0.9993 (0.9515)	0.52011 (0.51965)	0.84352 (0.83191)
$\mathbb{Z}_3$ covers	TW_2	1.9369 (1.9640)	0.51450 (0.51502)	0.96793 (0.96937)
$\mathbb{Z}_4$ covers	$\max(TW_1, TW_2)$	0.8811 (0.8099)	0.53763 (0.53895)	0.82293 (0.80643)
$\mathbb{Z}_5$ covers	$\max(TW_2, TW_2')$	1.6245 (1.6219)	0.52122 (0.52193)	0.94011 (0.93968)
quaternion covers	TW_4	3.1263 (3.2061)	0.51094 (0.51107)	0.99825 (0.99857)
K_5 covers	TW_1	0.9551 (0.9515)	0.51891 (0.51965)	0.83281 (0.83191)
$K_5 - e$ covers	TW_1	0.9387 (0.9515)	0.51890 (0.51965)	0.82916 (0.83191)
$K_4 - e$ covers	TW_1	0.7326 (0.9515)	0.52518 (0.51965)	0.79490 (0.83191)

TABLE 1. Observed statistics at the largest available size of each family against the values predicted by Conjectures 2, 3, and 4: the ratio $g_V = (\rho - \mu_V) / \sigma_V$ of the mean gap to the standard deviation, the mass left of the sample mean, and the fraction of samples below the Ramanujan threshold ρ ($\rho = 2\sqrt{3}$ except for the $K_5 - e$ and $K_4 - e$ rows, where ρ is the spectral radius of the universal cover of the base). Predicted values for the maximum laws were computed numerically from the Tracy-Widom distributions assuming independent blocks on a common scale. Each cell shows the observed value with the predicted limit in parentheses. Observed values are at $V = 500,000$ (simple), $n = 5000$ (abelian and quaternion covers), and $V = 10,000$ (the fixed base families). The remaining gaps in the Ramanujan column are the $1/n$ shifts of Table 2.

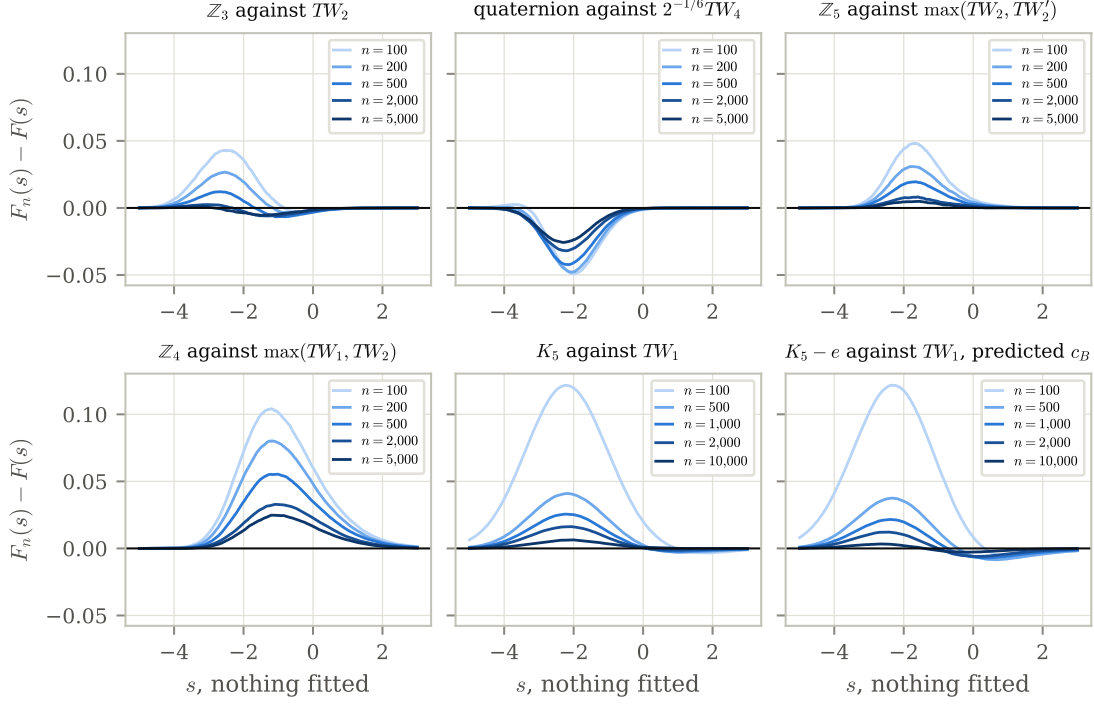


FIGURE 4.9. Parameter-free comparison of six cover families with their conjectured laws: $F_n(s) - F(s)$, where F_n is the empirical distribution function of $s = (\lambda - \rho)N^{2/3}/c$ with the predicted c , at five sizes each (lightest: smallest). The $K_5 - e$ panel uses the universal cover constant $C_B^{-2/3} = 0.695228$ of (2.4). Nothing is fitted.

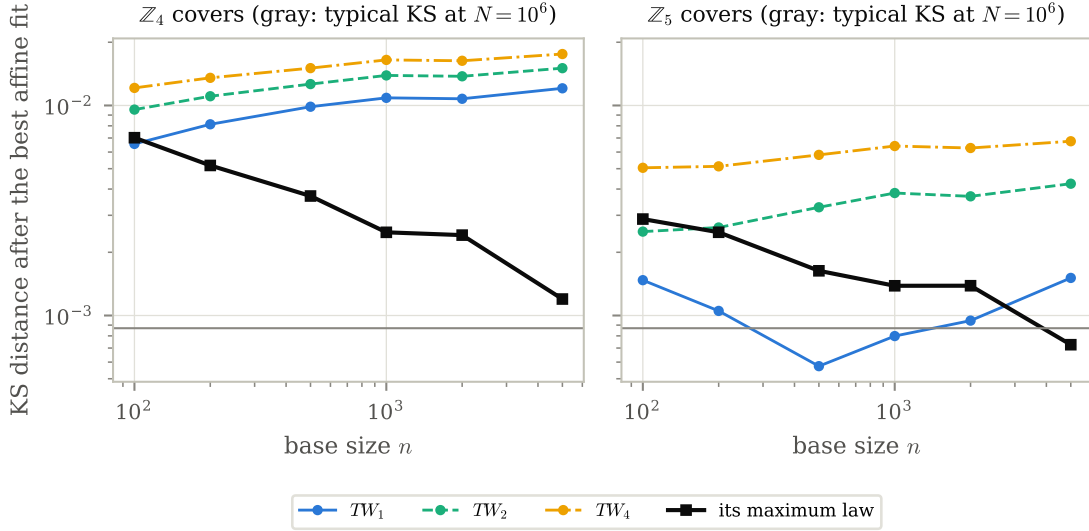


FIGURE 4.10. Kolmogorov-Smirnov distance after the best affine fit, for the $\mathbb{Z}_4$ (left) and $\mathbb{Z}_5$ (right) cover families, to the three Tracy-Widom laws and to the family's own maximum law, $\max(TW_1, TW_2)$ and $\max(TW_2, TW_2')$. Only the distance to the maximum law falls with n .

family	size	predicted scale	scale ratio γ	location a	KS after affine fit
simple 4-regular	$V = 20,000$	$c(4) = 0.832683$	1.0031 ± 0.0003	4.453 ± 0.020	0.7×10^{-3}
simple (legacy)	$V = 500,000$	$c(4)$	1.0036 ± 0.0033	4.30 ± 0.37	1.2×10^{-3}
multigraph	$V = 500,000$	$c(4)$	0.9992 ± 0.0027	-1.31 ± 0.34	1.6×10^{-3}
$\mathbb{Z}_3$ covers	$n = 5000$	$c(4)$	1.0083 ± 0.0006	-0.351 ± 0.018	0.8×10^{-3}
$\mathbb{Z}_4$ covers	$n = 5000$	$c(4)$	0.9901 ± 0.0012	1.022 ± 0.026	1.2×10^{-3}
$\mathbb{Z}_5$ covers	$n = 5000$	$c(4)$	1.0055 ± 0.0013	0.022 ± 0.028	0.7×10^{-3}
quaternion covers	$n = 5000$	$c(4), \chi_4 = 2^{-1/6}TW_4$	1.0052 ± 0.0008	-0.742 ± 0.026	0.5×10^{-3}
K_5 covers	$V = 10,000$	$c(4)$	1.0091 ± 0.0003	0.073 ± 0.012	0.7×10^{-3}
$K_5 - e$ covers	$V = 10,000$	$C_B^{-2/3} = 0.695228$	1.0119 ± 0.0003	-0.233 ± 0.014	0.6×10^{-3}
$K_4 - e$ covers	$V = 10,000$	$C_B^{-2/3} = 0.212519$	1.0441 ± 0.0009	-0.922 ± 0.007	2.8×10^{-3}

TABLE 2. The finite size law (2.6) at the largest size of each family: the predicted scale, the scale ratio, the location coefficient, and the Kolmogorov-Smirnov distance of the sample to its conjectured law after the affine fit, with standard errors from ten batches. The typical distance of a sample from its own law is 0.4×10^{-3} at 5×10^6 samples, 0.9×10^{-3} at 10^6 , and 2.7×10^{-3} at 10^5 .

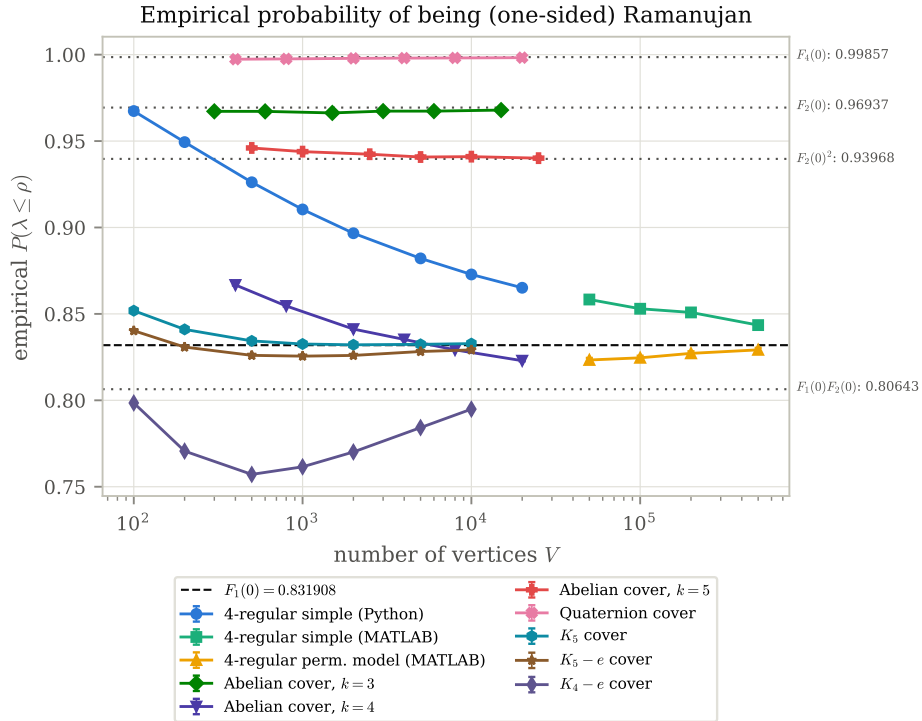


FIGURE 4.11. Empirical $P(\lambda \leq \rho)$ against V for all families, where ρ is the family's Ramanujan threshold, with the limit $F_1(0) = 0.8319081\dots$ for the $\beta = 1$ families. Each cover family approaches the product of Tracy-Widom functionals of Conjecture 3 listed in Table 1.

5. COVERS OF A FIXED BASE GRAPH

We now fix the base graph and let only the cover vary, testing three bases: the complete graph K_5 , whose covers are 4-regular and calibrate the fixed base model against section 3, the irregular graph $K_5 - e$, and the cycle rank two graph $K_4 - e$

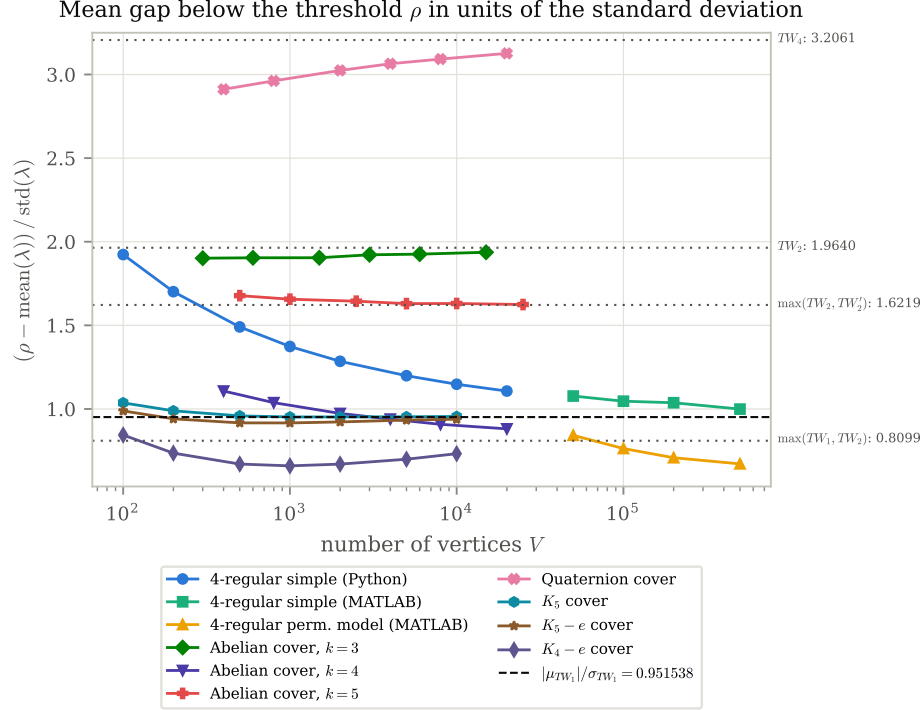


FIGURE 4.12. The ratio $g_V = (\rho - \mu_V) / \sigma_V$ against V for all families, where ρ is the family's Ramanujan threshold, with the $\beta = 1$ reference $|\mu_{TW_1}| / \sigma_{TW_1} = 0.951537\dots$ dashed. MNS conjectured this ratio tends to 0, however each family plateaus at the ratio of its own limit law: 1.96 for $\mathbb{Z}_3$, 1.62 for $\mathbb{Z}_5$, 3.21 for quaternion covers, and the $\beta = 1$ reference for the regular and fixed base families.

of subsection 5.3. Nothing in Conjecture 4 requires the base graph to be regular, and in the irregular case nothing is proven: the largest new eigenvalue should concentrate at the spectral radius $\rho(B)$ of the universal cover, an algebraic number lying between $2\sqrt{\delta} - 1$ and $2\sqrt{\Delta} - 1$, where δ and Δ are the minimum and maximum degrees of B , with TW_1 fluctuations on the scale $(C_B n k)^{-2/3}$ and limiting Ramanujan probability $F_1(0)$, universally in the base graph. The threshold $\rho(B)$ is also the benchmark of the X -Ramanujan property of Mohanty and O'Donnell [84], who construct infinite families of finite graphs whose second eigenvalue is at most the spectral radius of a fixed infinite cover. It is the right benchmark: by the generalized Alon-Boppana theorems of Friedman and Tillich [43], no infinite family of graphs sharing the universal cover of B has second eigenvalue asymptotically below $\rho(B)$, and Hoory [56] bounds $\rho(B)$ itself below by $2\sqrt{\bar{d}} - 1$, where $\bar{d}$ is the average degree of B . Random bipartite biregular graphs, whose limiting tree is not regular either, satisfy an Alon type theorem as well [24].

The main base is $K_5 - e$, the complete graph on five vertices minus an edge, one of the smallest irregular base graphs of minimum degree three, with eigenvalues $1 + \sqrt{7} = 3.6457\dots$, 0 , -1 , -1 , and $1 - \sqrt{7}$. Since the largest old eigenvalue exceeds $\rho(K_5 - e)$, the largest new eigenvalue of a cover is not even its second largest eigenvalue, so that the exact removal of the lifted base spectrum matters. By a result of McKay [81],

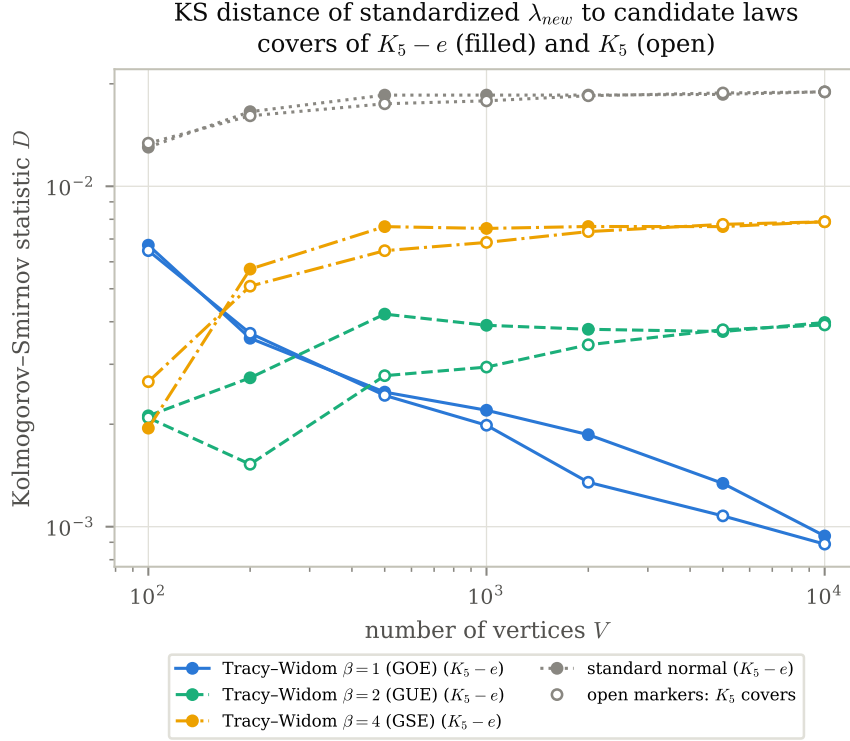


FIGURE 5.1. Kolmogorov-Smirnov distance of the standardized largest new eigenvalue to the four candidate laws for covers of $K_5 - e$ (filled markers, standardized around $\rho(K_5 - e)$) and of K_5 (open markers), 5×10^6 samples per size. For both families only the distance to $\beta = 1$ keeps falling.

$\rho(K_5 - e)$ is the largest zero of

$$x^{14} - 20x^{12} + 122x^{10} - 152x^8 - 1295x^6 + 4540x^4 - 5948x^2 - 2000,$$

an irreducible polynomial whose roots admit no expression in radicals, giving

$$\rho(K_5 - e) = 3.26287646593635862827 \dots,$$

which we re-derived and verified independently, and which also follows from the variational formula of Garza-Vargas and Kulkarni [46]. Note the contrast with the regular case, where $\rho = 2\sqrt{d-1}$: for a general base even the centering of the conjectured law is a non-trivial algebraic quantity (see [12, 46, 81] on computing ρ , and [31] for related spectral experiments on structured random graphs), and so is the scale constant C_B of (2.4).

5.1. The Bulk Law. We sampled the K_5 and $K_5 - e$ families at the scale of the strongest regular graph data: 5×10^6 independent uniform covers per size, at cover degrees $20 \leq k \leq 2000$, giving graphs on $V = 5k$ between 100 and 10,000 vertices. The covers of K_5 are 4-regular graphs with Ramanujan threshold $2\sqrt{3}$, of a different fine structure than the permutation model, since the vertex set carries a fixed partition into five independent fibers with edges only between fibers adjacent in K_5 , and so they test whether the limit law and its constants depend on anything beyond the degree. The covers of $K_5 - e$ test Conjecture 4 where no theorem reaches, around the algebraic centering $\rho(K_5 - e)$ and on the scale set by the new constant $C_{K_5 - e}$.

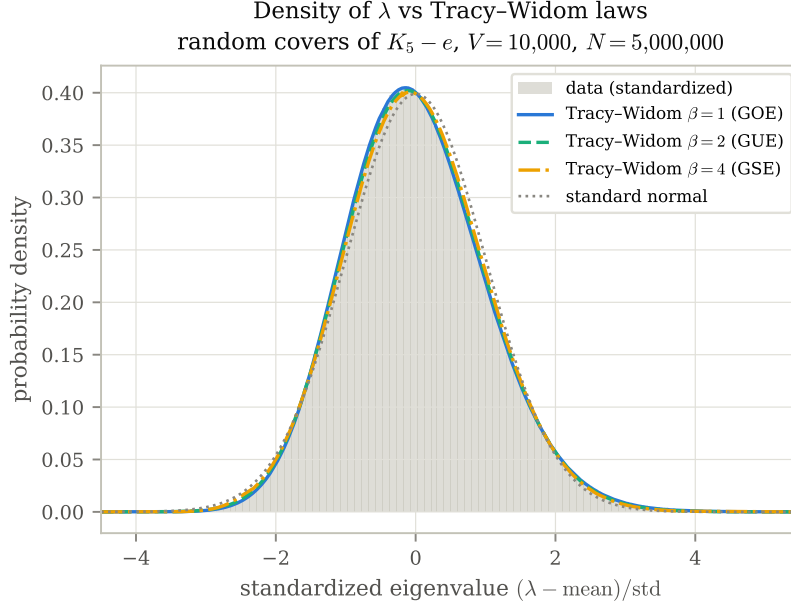


FIGURE 5.2. Standardized empirical density of the largest new eigenvalue over 5×10^6 random covers of $K_5 - e$ at $V = 10,000$, against the standardized Tracy-Widom densities and the standard normal.

Figure 5.1 shows the outcome of the distributional test. For both families the Kolmogorov-Smirnov distance to $\widetilde{TW}_1$ falls monotonically with V , reaching 8.9×10^{-4} for K_5 and 9.4×10^{-4} for $K_5 - e$ at $V = 10,000$, a small multiple of the sampling floor and equal to the simple 4-regular family's distance at the same size, while the distances to the other laws stall or grow, ending at 4.0×10^{-3} ($\widetilde{TW}_2$), 7.9×10^{-3} ($\widetilde{TW}_4$), and 1.9×10^{-2} (normal) for $K_5 - e$. A two sample comparison is sharper still: the Kolmogorov-Smirnov distance between the standardized simple and K_5 samples at $V = 10,000$ is 3.7×10^{-4} ($p = 0.88$), while each lies 8.9×10^{-4} from $\widetilde{TW}_1$ ($p \approx 10^{-3}$), and so at 10^7 combined samples the two models are statistically indistinguishable from one another in shape while both still differ measurably from the limit. At small sizes the families are well separated ($p = 2 \times 10^{-15}$ at $V = 100$), merging from $V \approx 5000$ on. Figure 5.2 shows the standardized density of the largest $K_5 - e$ dataset, and Figure 5.3 the corresponding quantile plot. The mass left of the mean reaches 0.51891 (K_5) and 0.51890 ($K_5 - e$) against the TW_1 value 0.519652, on the same convergence path as the simple family at equal sizes, and the ratio g_V reaches 0.9551 and 0.9387 against the TW_1 ratio $|\mu_{TW_1}|/\sigma_{TW_1}$ (Figure 4.12). The local exponents of the mean gap and standard deviation descend to -0.673 and -0.676 for K_5 and -0.670 and -0.678 for $K_5 - e$ over the last pair of sizes, approaching $-2/3$ as for the regular families (Figure 3.11). The empirical Ramanujan probabilities at $V = 10,000$ are

$$P(\lambda_{\text{new}} \leq 2\sqrt{3}) = 0.83281, \quad P(\lambda_{\text{new}} \leq \rho(K_5 - e)) = 0.82916,$$

against the conjectured $F_1(0) = 0.8319081\dots$ (Table 1). Every statistic selects the $\beta = 1$ class, as the reality of the standard representation predicts, and for $K_5 - e$ it does so around a centering which is not $2\sqrt{d-1}$ for any d .

The amplitudes are the sharpest test of Conjecture 4, since for $K_5 - e$ the predicted constant $C_B^{-2/3} = 0.695228\dots$ of (2.4) was computed from the universal cover alone,

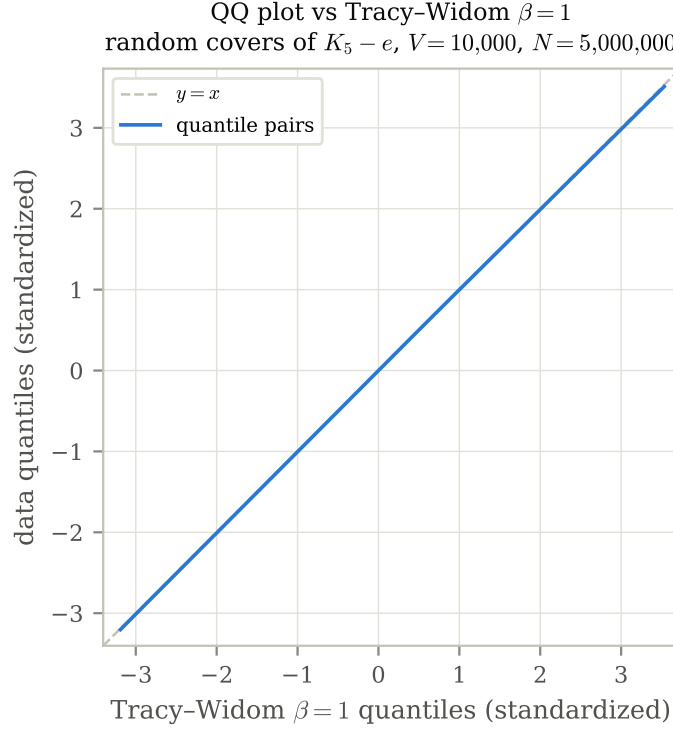


FIGURE 5.3. Quantile-quantile plot of the largest $K_5 - e$ dataset against $\widetilde{TW}_1$, probability range 10^{-4} to 0.999: the bulk of the law around the algebraic centering $\rho(K_5 - e)$ is Tracy-Widom at quantile resolution.

with no reference to the data. The fitted amplitudes $\gamma_V c(4)$ descend through 0.7209, 0.7085, and 0.7035 at $V = 2000$, 5000, and 10,000, and a fit $c + bV^{-2/3}$ over $V \geq 1000$ or $V \geq 2000$ extrapolates to 0.6943 to 0.6949, within 0.15% of the prediction (Figure 5.4). The same fits for K_5 give 0.8315 to 0.8319 against $c(4) = 0.832683$, and so the amplitude constant for covers of K_5 is the regular graph constant $c(4)$ itself, in the convention with V the cover size, as the Kesten-McKay law of μ_{K_5} requires. The speed of convergence differs, however: the K_5 family sits within 10^{-3} of the limiting Ramanujan probability from $V = 1000$ onward (0.83258 at $V = 1000$), while the simple 4-regular series is still at 0.87282 at $V = 10,000$, and the K_5 ratio g_V is already 0.9522 at $V = 1000$ against the simple series' 1.148 at ten times that size. The difference lies in the location coefficient, which is $a \approx 0.03$ for K_5 against 4.47 for the simple graphs (Table 2), and not in the scale, which for K_5 converges if anything more slowly, with scale ratio 1.046 at $V = 1000$ against 1.027. The Ramanujan probability does not see the scale, only the shift.

5.2. Localized Defects and the Far Tails. The third and fourth moments tell a different story, and the accounting is instructive. The excess kurtosis of the $K_5 - e$ family grows with V , from 0.26 at $V = 100$ to 2.52 at $V = 10,000$ against the TW_1 value 0.165, and its skewness drifts upward through 0.361 against 0.293, while the same statistics for K_5 are 0.28 and 0.292. Figure 5.7 locates the cause in a thin tail of far outliers, at frequencies near 10^{-5} : at $V = 10,000$ a fraction 9×10^{-6} of the $K_5 - e$ samples lie above $\rho + 10\sigma_V$, while the extreme sample of the simple 4-regular family sits only 9 standard deviations out. These outliers are the tangles of [38] made visible sample

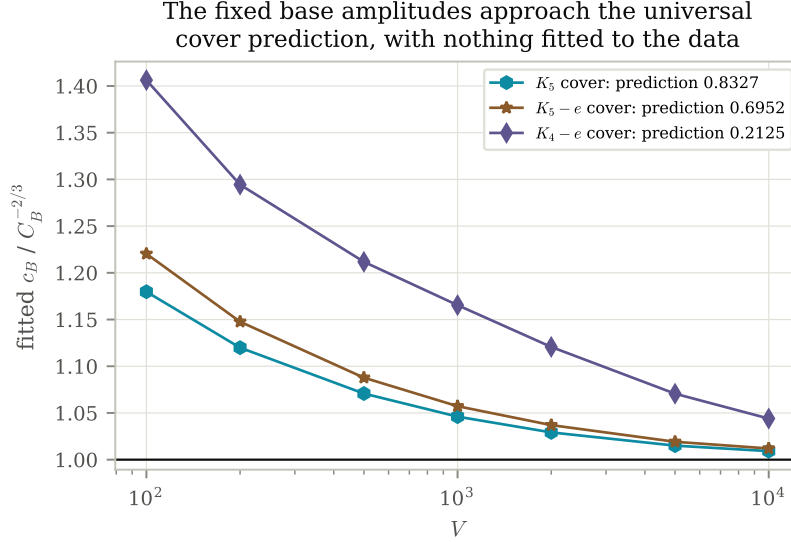


FIGURE 5.4. The fitted amplitudes of the three fixed base families over the universal cover predictions $C_B^{-2/3}$ of (2.4), none of which was fitted to the data. K_5 and $K_5 - e$ converge to 1 together, while $K_4 - e$, whose square root regime is narrow, converges more slowly.

by sample, and they admit a complete quantitative description, one half of which we can prove. For regular bases, Addario-Berry and Griffiths [1] showed that large new eigenvalues of random lifts are caused by small induced subgraphs, up to a constant factor, and the propositions below make this exact for the rate and the location of each defect.

Note that every subgraph of a cover projects to the base graph by a locally injective map, since distinct neighbors of a vertex of a cover lie over distinct neighbors of its projection. With this in mind, fix a connected graph H of minimum degree at least two admitting a locally injective map, an *immersion*, $\varphi : H \rightarrow B$, and write $r(H) = e(H) - v(H) + 1$ for its cycle rank. Let H^* be the infinite graph obtained from H by attaching, at each vertex x of H and each neighbor of $\varphi(x)$ not used by an edge of H at x , the corresponding branch of the universal cover of B , and define the *defect eigenvalue*

$$\lambda_B(H) = \|A_{H^*}\|,$$

the spectral norm of the adjacency operator of H^* , maximized over the locally injective maps φ when there are several. When B is a bouquet, H^* is the tree extension used by Dong and McKenzie [33] in their proof that every sequence $d \geq \mu_2 \geq \dots \geq \mu_k \geq 2\sqrt{d-1}$ is a limit of top eigenvalues of d -regular graphs. Deep inside any branch, H^* contains arbitrarily large balls of the universal cover, and so $\lambda_B(H) \geq \rho(B)$ always, and when the inequality is strict we say that H *binds* an eigenvalue above $\rho(B)$. When it binds, $\lambda_B(H)$ lies above the essential spectrum, since removing finitely many vertices from H^* leaves a union of branches, and $\lambda_B(H)$ is computable to any precision from the branch values (2.3) at real $\lambda > \rho(B)$: each attached branch contributes its value to the diagonal, and $\lambda_B(H)$ is the largest λ with $\lambda_{\max}(A_H + D(\lambda)) = \lambda$, where $D(\lambda)$ is the diagonal matrix of attached branch values (for eigenvalues of graphs with attached infinite tails see also [47]). Figure 5.5 shows the leading defects of our families. Two elementary facts organize the far tails: copies of H appear in a random cover at a

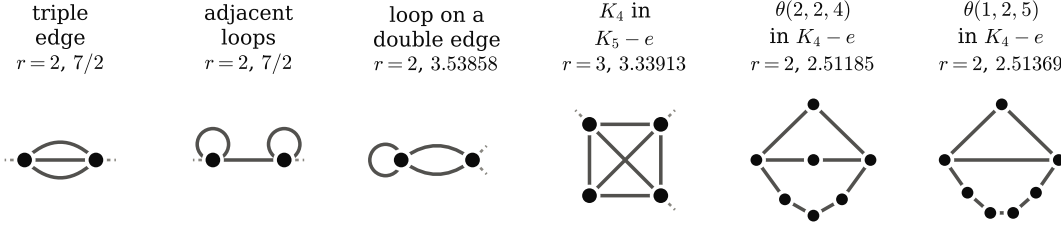


FIGURE 5.5. The leading defects, with their cycle ranks r and defect eigenvalues $\lambda_B(H)$: the three binding defects of cycle rank two of the permutation multigraph (subsection 3.4), K_4 immersed in $K_5 - e$ (in K_5 it has a fourth attached branch and binds at $7/2$), and the two shortest binding generalized thetas of $K_4 - e$ (subsection 5.3, branches not drawn). Dashed stubs are attached branches of the universal cover.

polynomial rate governed by the cycle rank, and any copy forces a new eigenvalue near $\lambda_B(H)$.

Proposition 5. *Let B be a connected base graph, let H be a connected graph of minimum degree at least two admitting a locally injective map to B , and let X_H be the number of subgraphs of a uniformly random degree k cover H_k of B isomorphic to H . Then*

$$E[X_H] = c_H k^{1-r(H)} (1 + O(k^{-1})), \quad c_H = \frac{J(H, B)}{|\text{Aut}(H)|},$$

where $J(H, B)$ is the number of locally injective homomorphisms from H to B , and, when $r(H) \geq 2$,

$$P(X_H \geq 1) = c_H k^{1-r(H)} (1 + o(1)).$$

Proof. Suppose first that B is simple, as are all of our fixed bases, since loops and multiple edges require only notational changes. Copies of H in the cover correspond to pairs consisting of a locally injective homomorphism $\psi : H \rightarrow B$ and a lift of ψ , with each copy counted $|\text{Aut}(H)|$ times. Fix ψ , and choose for each vertex of H a lift in the fiber over its image, injectively within each fiber, which can be done in $k^{v(H)} (1 + O(k^{-1}))$ ways. The chosen vertices span a copy of H exactly when, for each edge e of B , the permutation σ_e maps every specified source to its specified target. If m_e edges of H lie over e , then the sources are distinct and the targets are distinct, by fiber injectivity and the local injectivity of ψ , and so this event has probability $(k - m_e)!/k! = k^{-m_e} (1 + O(k^{-1}))$, independently across the edges of B . Multiplying and summing over ψ gives the expectation, since $v(H) - e(H) = 1 - r(H)$.

For the probability, we have that $E[X_H^2] = E[X_H] + E[Y]$, where Y counts ordered pairs of distinct copies. The union of two distinct copies is either disconnected, and such pairs contribute $c_H^2 k^{2-2r} (1 + O(k^{-1}))$ by the computation above, or connected of cycle rank at least $r+1$: if two distinct copies overlap, their intersection is a proper subgraph F of H , and $r(U) = 2r(H) - 1 - r(F) + q$ for the union U , where q is the number of components of F and $r(F) = e(F) - v(F) + q$, so it suffices that $r(F) \leq r(H) - 2 + q$. For connected F , with $q = 1$, this says $r(F) \leq r(H) - 1$, which holds since a proper connected subgraph of a graph of minimum degree two has strictly smaller cycle rank: a path leaving the subgraph cannot terminate, and closes an independent cycle whether it returns to the subgraph or to itself. For disconnected F , with $q \geq 2$, it follows from $r(F) \leq r(H)$, since the cycle spaces of disjoint subgraphs embed independently into

that of H . Each connected union type of cycle rank at least $r + 1$ contributes $O(k^{-r})$, and there are finitely many, and so $E[X_H^2] = E[X_H] (1 + o(1)) + E[X_H]^2 (1 + o(1))$. The claim follows from $P(X_H \geq 1) \geq E[X_H]^2 / E[X_H^2]$ and $P(X_H \geq 1) \leq E[X_H]$. $\square$

The second fact is that every occurrence produces its eigenvalue.

Proposition 6. *In the setting of Proposition 5, suppose that $r(H) \geq 2$ and $\lambda_B(H) > \rho(B)$. Then for every $\epsilon > 0$ there is a constant $c = c(B, H, \epsilon) > 0$ such that for all k sufficiently large,*

$$P(\lambda_{\text{new}}(H_k) \geq \lambda_B(H) - \epsilon) \geq c k^{1-r(H)}.$$

Proof. Let H_ℓ^* denote the truncation of H^* at depth ℓ from H . Since H^* is connected and locally finite, $\lambda_{\max}(H_\ell^*)$ increases to $\|A_{H^*}\|$ as $\ell \rightarrow \infty$, and so we may fix ℓ with $\lambda_{\max}(H_\ell^*) > \lambda_B(H) - \epsilon/2$. Let $u \geq 0$ be the Perron eigenvector of H_ℓ^* with $\|u\| = 1$, supported on $N = N(\ell)$ vertices. Call a copy of H in the cover *clean* if the depth ℓ neighborhood of its vertex set is isomorphic, over B , to H_ℓ^* . A copy which is not clean lies in a connected subgraph of the cover of size $O_\ell(1)$ and cycle rank at least $r(H) + 1$, since any deviation of the neighborhood from H_ℓ^* closes an additional independent cycle within distance ℓ , and the expected number of these is $O_\ell(k^{-r(H)})$ by the computation of Proposition 5, and so with probability at least $c_H k^{1-r(H)} (1 + o(1))$ the cover contains a clean copy.

Given a clean copy, transplant u onto its neighborhood. The subspace of functions which are constant on every fiber is invariant under the adjacency matrix A of the cover, its spectrum on this subspace is the spectrum of B , and its spectrum on the orthogonal complement is exactly the new spectrum. Writing P for the orthogonal projection onto the fiber constant functions, each fiber meets the support of u in at most N vertices, and so by the Cauchy-Schwarz inequality

$$\|Pu\|^2 = \sum_{b \in V(B)} \frac{1}{k} \left(\sum_{x \in \pi^{-1}(b)} u(x) \right)^2 \leq \frac{N}{k}.$$

Set $w = u - Pu$, a vector in the new subspace with $\|w\|^2 \geq 1 - N/k$. Since $u \geq 0$ and the neighborhood of the clean copy contains H_ℓ^* as a subgraph, we have that $\langle u, Au \rangle \geq \lambda_{\max}(H_\ell^*)$, and since $\|A\|$ is at most the maximum degree of B ,

$$\lambda_{\text{new}}(H_k) \geq \frac{\langle w, Aw \rangle}{\|w\|^2} \geq \langle u, Au \rangle - 3 \|A\| \|Pu\| \geq \lambda_{\max}(H_\ell^*) - O_\ell(k^{-1/2}),$$

whose right hand side is positive and exceeds $\lambda_B(H) - \epsilon$ for k large, and the result follows. $\square$

One family of defects is degenerate. A graph which is itself a cover of B , beginning with B under the identity, saturates every direction at every vertex, and so a copy of it is a full connected component, with defect eigenvalue its own Perron eigenvalue: its occurrence disconnects the cover, and by removal by value (subsection 2.4) the pipeline records instead the largest remaining eigenvalue outside the base spectrum. We exclude covers of B from the defect catalogue and from the minimum in the conjecture below, and by Proposition 5 the disconnected covers have probability $O(k^{1-r(B)})$. We believe that the remaining defects account for the entire polynomial tail, and conjecture the matching upper bound.

Conjecture 7. *Let B be a connected base graph of cycle rank at least two, and let H_k be a uniformly random degree k cover of B , conditioned on connectedness. Then for every $\epsilon > 0$,*

$$P(\lambda_{\text{new}}(H_k) > \rho(B) + \epsilon) = k^{1-r(\epsilon)+o(1)},$$

where

$$r(\epsilon) = \min \{r(H) : H \text{ immerses in } B \text{ and is not a cover of } B, \lambda_B(H) > \rho(B) + \epsilon\},$$

and where the probability is smaller than every power of k when no such H exists.

For regular Ramanujan bases, Conjecture 7 is essentially the theorem of Friedman and Kohler [42, 41], and the coexistence of a Tracy-Widom bulk with a thin family of localized outliers is the picture proven at the sparse Erdős-Rényi edge [7, 8, 9, 55], and for sparse Wigner matrices the outliers are likewise produced by localized defects [100]. The eigenvectors near a Tracy-Widom edge are delocalized, and for random regular graphs they are locally Gaussian waves [51], as they are in small spectral windows for random lifts [32], so that the two populations differ in their eigenvectors as well as in their eigenvalues. The mechanism is not tied to randomness: Alon, Ganguly, and Srivastava [5] construct regular graphs of high girth whose top eigenvectors are fully localized on small vertex sets, and every value in $[2\sqrt{d-1}, d]$ is a limit point of second eigenvalues of d -regular graphs [6, 33]. Proposition 6 is the elementary half in general. Together with the fluctuation conjecture, the defect picture determines exactly which statistics converge.

Corollary 8. *Assume the scaling of Conjecture 4, and suppose that some H immersing in B , not a cover of B , has $\lambda_B(H) > \rho(B)$, with $r^* \geq 2$ the least cycle rank of such an H . Then the p^{th} absolute moment of $(\lambda_{\text{new}}(H_k) - \rho(B)) k^{2/3}$ diverges as $k \rightarrow \infty$ for every $p > 3(r^* - 1)/2$.*

To see this, note that a defect at fixed height $\Delta = \lambda_B(H) - \rho(B)$ sits at $\Delta/\sigma_k \asymp \Delta k^{2/3}$ standardized units above the mean, and occurs with probability $\asymp k^{1-r^*}$ by Propositions 5 and 6, and so contributes to the p^{th} standardized moment at order

$$(5.1) \quad k^{1-r^*} (\Delta k^{2/3})^p \asymp k^{1-r^* + \frac{2p}{3}},$$

which diverges exactly when $p > 3(r^* - 1)/2$. Under Conjecture 7, holding uniformly down to the fluctuation scale, the moments below the threshold converge instead to those of the limit law. Equation (5.1) organizes every moment anomaly in this paper. For the permutation multigraph model and for $K_4 - e$, where $r^* = 2$, the variance itself is eventually dominated by the defect band, growing relative to the bulk like $k^{1/3}$, and at the bulk scale the skewness diverges like k and the kurtosis like $k^{5/3}$, so that the raw moments of these families are the most contaminated in the dataset, which is what subsection 3.4 observed, and the raw ratio g_V of subsection 5.3 must eventually turn around and decrease, with only its trimmed version converging to the TW_1 ratio. For $K_5 - e$ and K_5 , where $r^* = 3$, the variance is safe, the kurtosis diverges like $k^{2/3}$, matching the observed growth, and $p = 3$ is exactly critical: the defects add a constant, non-vanishing offset to the skewness, which is why the $K_5 - e$ skewness settles near 0.36 rather than at the Tracy-Widom value 0.293, while for K_5 the constant is too small to resolve. Every quantile level statistic is immune, since the total defect mass vanishes: the Kolmogorov-Smirnov distance moves by at most the tail mass and the Ramanujan probability by less, and this is why the fixed base families are omitted from Figure 4.5 while every distributional test of subsection 5.1 converges.

base B	defect H	$r(H)$	$\lambda_B(H)$	rate k^{1-r}	fitted exponent
bouquet B_2	triple edge, adjacent loops	2	7/2	V^{-1}	too few to fit
bouquet B_2	loop on a double edge	2	3.5385847...	V^{-1}	too few to fit
bouquet B_2	triangle with a loop	2	3.4675039...	V^{-1}	too few to fit
$K_5 - e$	K_4	3	3.3391340785...	k^{-2}	2.02 ± 0.13
$K_5 - e$	largest rank four	4	3.4519402...	k^{-3}	2.90 ± 0.14
K_5	K_4	3	7/2	k^{-2}	1.7 ± 0.7
$K_4 - e$	$\theta(2, 2, 4)$	2	2.5118521...	k^{-1}	≈ 1.3 , descending
$K_4 - e$	$\theta(1, 2, 5)$	2	2.5136918...	k^{-1}	≈ 1.3 , descending
$K_4 - e$	rank three band	3	$\lesssim 2.530$	k^{-2}	1.96

TABLE 3. The catalogue of leading localized defects. Each row lists a subgraph H immersing in B , its cycle rank $r(H)$, the defect eigenvalue $\lambda_B(H)$ computed by the branch recursion, the occurrence rate $k^{1-r(H)}$ of Proposition 5, and the exponent fitted to the observed outlier counts. The bouquet rows are the permutation multigraph model of subsection 3.4, with $k = V$, where 18 samples across four sizes are too few for a fitted exponent. Here $\theta(a, b, c)$ is the generalized theta graph whose two branch vertices are joined by routes of lengths a , b , and c , and the two $K_4 - e$ atoms sit too close together for separate counts, so one exponent is fitted to the pair.

The data confirms the computed defects in detail. Evaluating $\lambda_B(H)$ by the branch recursion, and confirming the values by planting the subgraphs in sampled covers, we have that for $K_5 - e$ no subgraph of cycle rank two binds an eigenvalue above ρ , that K_4 , of cycle rank three, binds

$$\lambda_{K_5-e}(K_4) = 3.3391340785\dots,$$

with its subdivisions filling the interval below it, and that the largest cycle rank four value is 3.4519402..., while a disjoint copy of $K_5 - e$ itself would give $1 + \sqrt{7}$. The outlier values at the largest sizes fill an interval above ρ , leave a gap, and cluster on the K_4 atom, where the extreme samples at $V = 5000$ and 10,000 sit, and the extremes at $V \leq 2000$ sit instead on the rank four atom, which decays faster and is gone by $V = 5000$ (Figure 5.6). The frequencies obey Proposition 5: the count in a window around the K_4 atom decays with fitted exponent 2.02 ± 0.13 against the predicted k^{-2} , and at thresholds beyond the reach of cycle rank three the fitted exponent is 2.90 ± 0.14 against k^{-3} . The apparent lengthening of the tail, from 27 standard deviations at $V = 5000$ to 42 at $V = 10,000$, is then exact: a fixed atom measured in units of $\sigma_V \propto V^{-2/3}$ lengthens by $2^{2/3} = 1.587$, and the observed ratio is 1.584. For the K_5 family the same computation gives the atom

$$\lambda_{K_5}(K_4) = \frac{7}{2}$$

exactly, and the six samples more than nine standard deviations above the mean at $V = 10,000$ all lie in $[3.497, 3.510]$. Table 3 collects the catalogue across all four families with binding defects of low rank.

5.3. The Boundary Case $K_4 - e$. The third fixed base, $K_4 - e$, tests Conjecture 4 at its boundary: the complete graph on four vertices minus an edge has cycle rank exactly

Far outliers lock onto the localized defect eigenvalues (dotted); frequencies decay like k^{1-r}

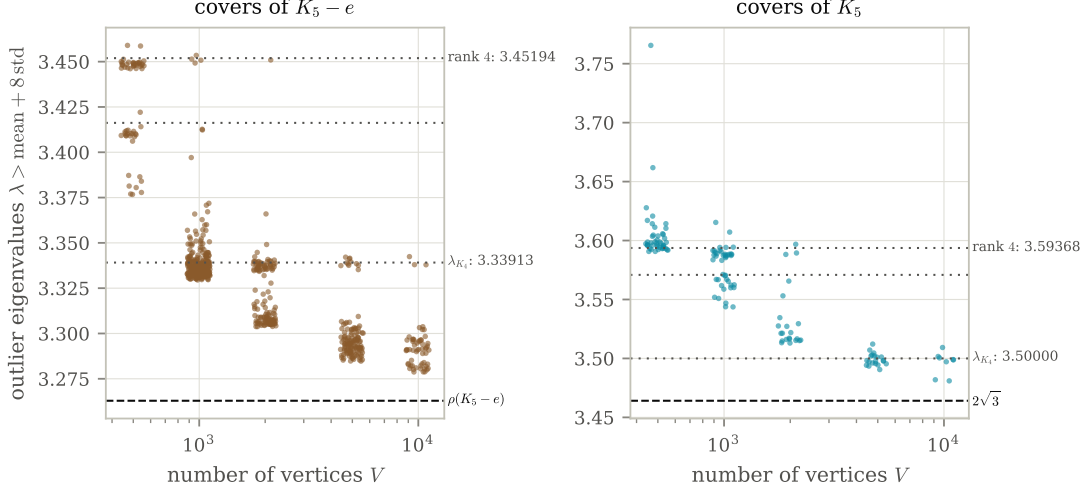


FIGURE 5.6. Every sample more than eight standard deviations above the mean, against V , for the $K_5 - e$ (left) and K_5 (right) cover families, with the computed defect eigenvalues overlaid: the outliers lock onto the K_4 atoms, $3.33913\dots$ for $K_5 - e$ and exactly $7/2$ for K_5 , and onto the cycle rank four values (dotted), while the continuum of K_4 subdivisions drains toward ρ and the frequencies decay like k^{1-r} .

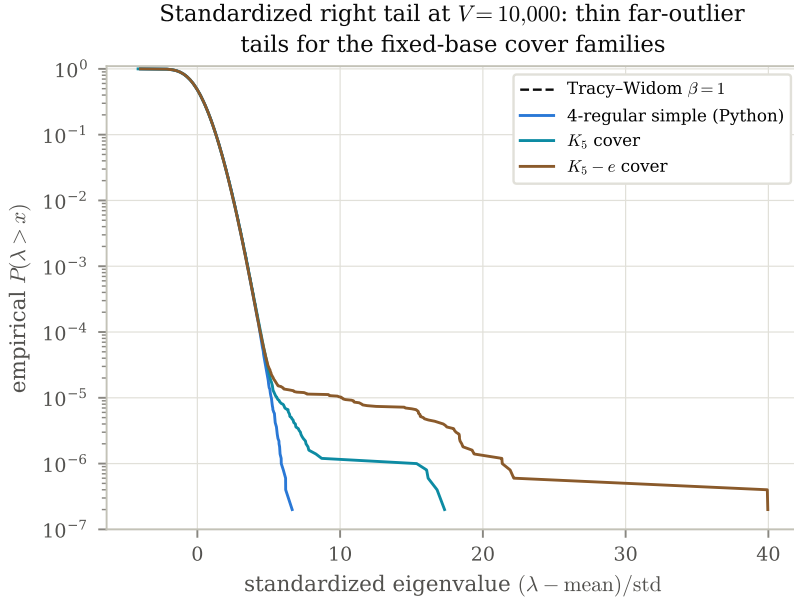


FIGURE 5.7. Standardized right tail $P(\lambda > x)$ at $V = 10,000$ for the simple 4-regular, K_5 cover, and $K_5 - e$ cover families. The bulk of all three collapses onto the Tracy-Widom $\beta = 1$ tail, while the $K_5 - e$ family carries a thin far outlier tail at frequency near 10^{-5} , reaching 42 standard deviations, and the K_5 family a much thinner one.

two, degrees $(2, 3, 3, 2)$, and the closed form

$$\rho(K_4 - e) = \sqrt{1 + 2\sqrt{7}} = 2.5082867902 \dots$$

Suppressing its two degree two vertices identifies $K_4 - e$ with the theta graph, two vertices joined by three parallel edges, and so random covers of $K_4 - e$ are subdivisions of random theta covers, which are exactly the bipartite cubic multigraphs built from three independent uniform permutations, the bipartite model of MNS and of Novikoff's experiments [89]. We sampled 10^6 covers per size at the same seven sizes $V = 4k$ between 100 and 10,000.

The family fits the framework, and it converges more slowly than any other. The mean approaches ρ from below with the ratio g_V rising through 0.7326 at $V = 10,000$ and the Ramanujan probability through 0.79490, both far from their conjectured limits $|\mu_{TW_1}|/\sigma_{TW_1}$ and $F_1(0)$. Two separate causes slow it down. The first is the scale. The square root edge of μ_{K_4-e} is steep, $C_B = 10.207$, and narrow: $\pi \mu'_B(x)/\sqrt{\rho - x}$ is within 3% of C_B only within about 10^{-3} of the edge, against about 10^{-2} for $K_5 - e$ (Figure 2.3), and the fluctuation scale $C_B^{-2/3}V^{-2/3}$ enters this window only at $V \approx 3000$. Accordingly the fitted amplitude descends slowly, through 0.2382, 0.2275, and 0.2219 at $V = 2000, 5000$, and 10,000, and extrapolates to 0.214 to 0.216 against the predicted $C_B^{-2/3} = 0.212519$, consistent with the prediction but with a larger uncertainty than for $K_5 - e$ (Figure 5.4). The second cause is the far tail. The raw distributional statistics are contaminated in a way the other covers only hint at: the excess kurtosis grows to 12.7, and the Kolmogorov-Smirnov distance of the standardized sample to $\widetilde{TW}_1$, after falling to 3.8×10^{-3} at $V = 500$, grows again to 1.9×10^{-2} at $V = 10,000$ (Figure 5.8, left). Removing the tail, by discarding the 0.2% of samples beyond six standard deviations, restores $\widetilde{TW}_1$ as the nearest law from $V = 1000$ on, with distance 4.0×10^{-3} at $V = 10,000$ against 6.4×10^{-3} and 9.9×10^{-3} for $\widetilde{TW}_2$ and $\widetilde{TW}_4$, and the affine quantile fit, which never sees the tail, gives 2.8×10^{-3} .

The tail itself differs from the $K_5 - e$ tail exactly as the defect picture predicts. No cycle rank one subgraph binds an eigenvalue above ρ , and no connected 2-cover of $K_4 - e$ has a new eigenvalue above ρ either, as a direct check of all 32 signings shows, however the base now has cycle rank two, and so proper rank two subgraphs exist, generalized thetas whose routes are longer than the base's and which carry branches, occurring at the rate k^{-1} of Proposition 5: the two shortest, $\theta(2, 2, 4)$ and $\theta(1, 2, 5)$ in the notation of Table 3, bind at

$$2.5118521 \dots \quad \text{and} \quad 2.5136918 \dots,$$

computed by the same branch recursion, one power of k more frequent than the leading $K_5 - e$ defects. Above the rank two band, rank three shapes reach ≈ 2.530 at rate k^{-2} . The data shows precisely this structure (Figure 5.8, right): the outliers form a dense band draining toward ρ rather than isolated atoms, capped near 2.530 at the largest sizes, with the count above the rank two cap decaying with fitted exponent 1.96 and the count at the cap decaying visibly more slowly, still descending toward the predicted 1. In the notation of Conjecture 7, $r(\epsilon) = 2$ for ϵ below ≈ 0.006 and 3 up to ≈ 0.022 . The comparison across the three fixed bases is the point: the denser the base, the higher the cycle rank at which the defect hierarchy starts, and the further the atoms sit from ρ . For $K_4 - e$ the hierarchy starts at rank two, the band crowds the bulk at accessible sizes, and the moments and even the raw Kolmogorov-Smirnov distance feel it, which

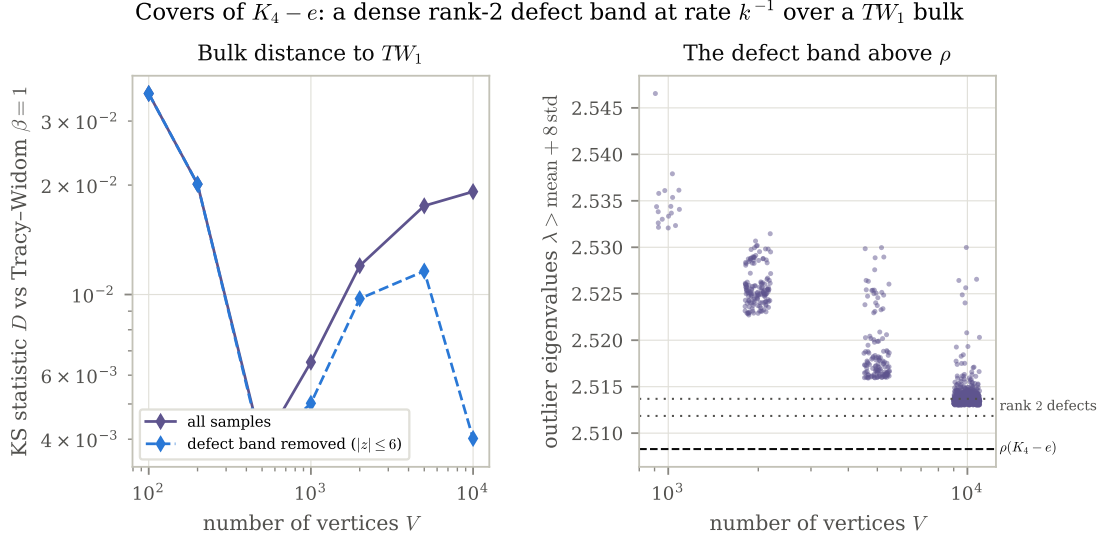


FIGURE 5.8. Covers of $K_4 - e$. Left: Kolmogorov-Smirnov distance of the standardized sample to $\widetilde{TW}_1$, for all samples and with the defect band removed. Right: every sample more than eight standard deviations above the mean against V , with $\rho(K_4 - e)$ dashed and the two computed rank two defect eigenvalues dotted. The band drains toward ρ at rate k^{-1} , with rank three shapes reaching ≈ 2.530 at rate k^{-2} .

together with its narrow square root edge is why this family, at the boundary of the conjecture, converges last.

6. OPEN PROBLEMS

The Block Laws. Conjecture 2 is open in every case, including $\beta = 1$: for covers of a d -regular base of size $n \geq 2$ no analog of Theorem 1 is known, and for irregular bases even the correct formulation of edge rigidity around $\rho(B)$ is open. The abelian case seems the natural first target, since the block A_χ is an explicit random Hermitian matrix built from a random regular graph and independent uniform phases, and the loop equation approach, which characterizes the Airy_β statistics for every rational β and has already given a simplified proof of Theorem 1 [23], seems well suited to it (see also [62]).

Problem 9. Prove that the largest eigenvalue of the $\mathbb{Z}_3$ -character block A_χ of a random $\mathbb{Z}_3$ -cover of a random simple d -regular graph has TW_2 fluctuations, in the sense of Conjecture 2.

The Scale and the Location. All of the scale constants in our data are predicted by the square root edge of the limiting spectral measure, with the Gaussian ensembles normalized alike, and the prediction is confirmed to within 0.15% after extrapolation for K_5 and $K_5 - e$. The location is a different matter. The shift a/n of (2.6) has a family dependent constant, from 4.47 for the simple 4-regular graphs to -1.44 for the multigraph and nearly 0 for the covers of K_5 , and it controls the approach of every Ramanujan probability to its limit, at rate $n^{-1/3}$.

Problem 10. Determine the location coefficient a . In particular, is a determined by the expected numbers of short cycles of the model, as the $1/n$ corrections to the

Kesten-McKay density computed by Metz, Parisi, and Leuzzi [82] suggest, and does the difference of about 6 between the simple and multigraph models come from the conditioning on the absence of loops and double edges?

I believe that the shift is a deterministic consequence of the short cycles, since in the families of our data it changes when the cycle statistics are changed and not otherwise, but I do not know how to compute it.

Far Outliers. The permutation multigraph model of subsection 3.4 has the Tracy-Widom bulk of the simple model together with a thin tail of defects of cycle rank two, and the fixed base families of section 5 carry tails of the same character.

Problem 11. Prove that the largest non-trivial eigenvalue of the permutation multigraph model, rescaled as in Theorem 1, converges to TW_1 in distribution. Is the point process of its eigenvalues above $2\sqrt{3} + \epsilon$ asymptotically a Poisson process on the three defect eigenvalues of subsection 3.4, with total intensity of order $1/V$?

Problem 12. Prove the upper bound in Conjecture 7: that nothing but the defects of subsection 5.2 inhabits the polynomial tail. The lower bound is Proposition 6, the regular Ramanujan case is the theorem of Friedman and Kohler [42, 41], and the fitted exponents of subsections 5.2 and 5.3 match the conjectured rates on two irregular bases with different leading ranks.

Growing Covers. Huang and Yau [61] proved Tracy-Widom fluctuations for regular graphs of growing degree. The analogous regime for covers, base size fixed and cover group growing, is precisely the setting of Conjecture 4, however one may also grow the base: for $\mathbb{Z}_k$ covers with k growing with n , the number of GUE blocks grows, and the largest new eigenvalue becomes the maximum of a growing number of Tracy-Widom variables. The maximum of independent Tracy-Widom variables lies in the Gumbel domain of attraction [29], which suggests a transition from Tracy-Widom to Gumbel fluctuations in the spirit of [64]. For iterated 2-covers of dense random regular graphs, Liu and Zou [72] prove exactly this maximum of independent Tracy-Widom structure.

Problem 13. For random $\mathbb{Z}_k$ covers of a random d -regular graph on n vertices, determine the joint regimes of (k, n) in which the largest new eigenvalue has Tracy-Widom, and Gumbel, fluctuations.

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